Dendroaspis: Detecting Agent-Orchestrated Attacks on High-Entropy Workstations via eBPF and Mamba

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***Abstract*—Modern host telemetry — kernel-function arguments, container metadata, full process lineage, structured capability fields — has outpaced the public host-IDS datasets built for syscall-id traces. We release Dendroaspis, a Mamba-based host-anomaly detector and the 19.3-million-event Tetragon eBPF behavioral-telemetry corpus collected from a Linux developer environment, with 61 per-trial labeled attack intervals across 22 ATT&CK technique identifiers spanning the breadth of MITRE ATT&CK—initial access, execution, persistence, privilege escalation, defense evasion, credential access, discovery, command and control—including a synthetic npm-postinstall supply-chain capstone. The labeled campaign is agent-orchestrated: an LLM-assisted planning agent performed reconnaissance, technique selection, and machine-paced execution (median inter-trial cadence $5.6\,$s), distinguishing the corpus from manually-operated red-team setups.**

We use the corpus to evaluate selective state-space sequence modeling for unsupervised host-telemetry anomaly detection. Two Mamba-based scorers — next-event likelihood (NLL) and field-aware masked event modeling (MEM-FA) — reach per-event AUROC of 0.74 and 0.75 in a rich 42-feature encoder; a flat-encoder ablation that drops seven high-cardinality identity hashes lifts MEM-FA to 0.78. Both clear unsupervised classical baselines (Isolation Forest 0.67, n-gram 0.57) on the same features. At each model's preferred per-window aggregator, the unsupervised flat-encoder MEM-FA Mamba detector ($\text{median} = 0.880$) modestly exceeds a generous leave-one-trial-out XGBoost supervised oracle ($\text{mean} = 0.842$) by $+0.038$ AUROC.Field-aware decomposition recovers an inversion on the Unix-Shell technique (`T1059.004` — operator-spawned bash) that next-event likelihood alone cannot escape (AUROC $0.62 \rightarrow 0.75$ under flat encoding); the two objectives reach near-identical headline AUROC by mechanistically distinct routes (NLL sharp on a small set of techniques, MEM-FA flat across all of them).

We frame these results as evidence that selective state-space models are a credible foundation for behavioral host anomaly detection in unsupervised, edge-deployable settings, not as a state-of-the-art claim. The released dataset, code, campaign provenance, and per-trial documentation are intended to support broader research — on encoder design, aggregator design, and adaptation of the same paradigm to other device classes (Windows endpoints, macOS, mobile, IoT, industrial control / OT).

***Index Terms*—anomaly detection, state space models, Mamba, eBPF, host telemetry, masked event modeling, MITRE ATT&CK, agent-orchestrated attacks, developer workstation security**

I. Introduction

Motivation. Host-based intrusion detection has spent thirty years on the same paradigm: model the benign sequence of system-level events with an autoregressive sequence model, score events by their negative log-likelihood under the model, and flag the events the model finds surprising [1], [2]. Modern eBPF kernel observability [3] has made the input side of this paradigm vastly richer than the syscall-id traces of the 1990s — per-event records now carry kprobed kernel-function arguments, container metadata, full process lineage, and structured capability and credential fields. The selective state-space model class introduced by Mamba [4], [5] provides a body that scales linearly in sequence length and runs efficiently on edge hardware [6], [7], making the next-event paradigm tractable on continuous high-rate telemetry streams.

Public threat reports now document the weaponization of Large Language Models (LLMs) and Agentic AI systems to plan, automate, and execute sophisticated attacks, with at least one documented case of an AI Agent operating across nearly all MITRE ATT&CK tactics in active intrusions [8]. Concurrently, AI is erasing the technical barrier historically required to perform such attacks, which is directly contributing to the scale and effectiveness of such attacks—such as the Shai-Hulud npm worm, which targets developer toolchains for credential theft and persistence [9]—demonstrating that attackers can now execute multi-stage and effective campaigns at machine speed with minimal manual oversight.

This paper's contributions. We make three contributions:

1) **The Dendroaspis Corpus: a 19.3M-event eBPF host-telemetry corpus with a labeled, AI-agent-orchestrated, ATT&CK-aligned campaign.** [10]. 19,301,915 Tetragon/eBPF events from a single Proxmox/LXC host running representative IDE / language-server / package-manager / shell workloads, partitioned $\sim 88/12$ train/test by the campaign-window time cut. The test partition contains 61 labeled attack intervals across 22 ATT&CK techniques (full per-technique reference in Appendix A) — including a synthetic npm-postinstall supply chain compromise (Track A: `T1195.002`, 8 labeled trials, 15,703 events) modeled after the same threat class as the Shai-Hulud npm worm [9] but without its self-propagating or real exfiltration — plus a chained post-compromise scenario (`CHAIN.CAP01`: shell $\rightarrow$ discovery $\rightarrow$ credential read $\rightarrow$ loopback exfil $\rightarrow$ cleanup, 34,130 events). The campaign was structured through LLM-assisted operational planning

with reconnaissance-conditioned technique selection (e.g., OS-gated technique exclusion) and machine-paced execution (median inter-trial cadence 5.6 s; §V-F). The agent role is documented as a reconstruction; full transcripts deferred to future work (§VII-A).

2) **The DENDROASPIS Detector: a Mamba sequence detector with two objectives plus a post-hoc encoder ablation.** [11] Per-event encoded features fed into a Mamba state-space model trained under (a) next-event likelihood and (b) field-aware masked event modeling. NLL reaches per-event $\mathrm{AUROC} = 0.74$ and MEM-FA reaches 0.75 in the rich 42-feature encoder; a flat-encoder ablation that drops seven high-cardinality identity hashes lifts MEM-FA to 0.78. Classical baselines (Isolation Forest, n-gram) reach 0.57–0.67 on the same data.
3) **An empirical comparison of next-event likelihood and field-aware masked event modeling, plus an encoder $\times$ architecture decomposition.** Both objectives reach $\mathrm{AUROC} \approx 0.74$–$0.75$ by mechanistically distinct paths; field-aware decomposition recovers an inversion on the Unix-Shell technique (`T1059.004` — bash, 175,797 events) that next-event likelihood cannot escape, with the recovery amplified further under the flat-encoder ablation (AUROC $0.62 \rightarrow 0.75$). Window-level aggregation under $\max$ is pessimistic by 0.22–0.35 AUROC for the Mamba scorers; median is the correct default.At the matched-aggregator comparison, the unsupervised DENDROASPIS flat-encoder MEM-FA ($\mathrm{median} = 0.88$) exceeds the supervised XGBoost-LOOCV reference ($\mathrm{median} = 0.84$)across all primary aggregators.

Non-claims. We do not claim state-of-the-art on any benchmark; no host-telemetry benchmark with directly-comparable published numbers exists. We do not claim generalization beyond a single-host LXC corpus with a single developer baseline. We do not claim a fully reproducible closed-loop autonomous attack trace; the agent role is reconstructed from the campaign yaml, the labeled trial structure, the distro-gating evidence, and the inter-trial cadence (§III-G). We do not claim live-deployment performance; the evaluation is offline against a frozen test partition.

Roadmap. §II positions the contribution against five paradigms in the prior literature. §III specifies the two-stage pipeline, encoder, body, three detection heads (NLL, vanilla MEM control, field-aware MEM), the agent-orchestrated attack campaign and threat model, and the training procedure. §IV pre-registers the five experiments and their falsification criteria. §V reports the five pre-registered experiments, the per-technique structure, the per-event score distributions, the cadence measurement that anchors the threat model, and the honest negatives. §VI interprets the results, names the threats to internal, external, and construct validity, and bounds the scope. §VII restates the contribution and schedules future work.

II. RELATED WORK

The detector sits at the intersection of five paradigms. We map each paradigm, name what it cannot do that this paper does, and close with the paper's position relative to them.

Classical sequence anomaly on syscall traces. Forrest et al. [1] introduced short fixed-length sequences of system calls as a signature of normal process behavior, with detection by enumeration of unseen sequences against a benign self-database. Warrender et al. [12] compared four detectors (STIDE, t-STIDE, RIPPER, hidden Markov models) on the same task to answer the question: is the power coming from the model, or from the syscall sequence data itself? They established that simple n-gram enumeration performs comparably to HMMs at a fraction of the training cost, locating most of the detection signal in the syscall sequence itself rather than in the model class. Wagner and Soto [13] demonstrated mimicry attacks that evade n-gram self databases entirely by composing exploits exclusively from in-self subsequences, motivating richer probabilistic and feature-aware approaches that raise but do not eliminate the cost of evasion. *This paradigm operates on a closed syscall-id vocabulary; it has no representation for kprobed kernel functions, structured arguments, container metadata, or process lineage — the fields that carry attack-vs-developer-baseline signal in eBPF kernel telemetry.*

Neural log-key sequence modeling. Du et al. [2] reframed anomaly detection as next-event language modeling on log-key sequences, using an LSTM trained only on benign sequences and flagging events that fall outside the model's top-g predictions. Meng et al. [14] extended the recipe with semantic template embeddings and a parallel count-based channel, demonstrating that richer per-event features improve next-event scoring. Zhang et al. [15] introduced FastText template embeddings and a Bi-LSTM with attention to tolerate log-format drift. Le and Zhang [16] argued that log parsing is the dominant failure mode of these pipelines and applied a BERT encoder directly to raw log lines, eliminating the parser. Lv et al. [17] pivoted from next-event prediction to representation quality, contending that geometrically compact normal-cluster embeddings detect anomalies even when next-event likelihood is in-distribution. Dietz et al. [18] consolidate the recent trajectory in ContraLog with parser-free dual-encoder contrastive learning over continuous message embeddings. *Every system in this lineage targets text-style application or system logs (HDFS, BGL, OpenStack, Thunderbird) at vocabularies of* 300–1000 *log keys; none publishes a recipe for ingestion of structured kernel-eBPF event streams, where the per-event representation is a* 33*-column tuple over heterogeneous categorical, hashed, and continuous fields rather than a single tokenizable string.*

Masked sequence modeling and continuous-target SSL. Devlin et al. [19] introduced the masked-language-modeling objective in BERT: at training time, 15% of input positions are replaced with a learned `[MASK]` token and the model predicts the original token at masked positions via cross-entropy over a vocabulary. Guo et al. [20] adapted MLM to log anomaly de-

tection in LogBERT, demonstrating that masked-token reconstruction outperforms next-event NLL on log-key sequences. Baevski et al. [21] generalized masked modeling to continuous targets across speech, vision, and language by predicting the encoder's own contextualized representations under a stop-gradient EMA target, identifying continuous-target collapse as the dominant stability failure and exponentially-moving-average target encoders as the dominant mitigation. *We adopt the simplest possible instantiation of continuous-target MEM (vanilla detached mean-squared-error (MSE) in unnormalized embedding space; no EMA target, no L2-normalization of targets or predictions, no contrastive auxiliary) as a control, document its collapse on host-telemetry features, and report a field-aware redesign (per-field categorical and binary cross-entropy heads at masked positions) that recovers headline parity with next-event NLL — a structural fix to the vanilla-MEM failure mode rather than a hyperparameter swap.One architectural distinction from this lineage: our masked reconstruction runs on a causal Mamba body, not the bidirectional encoder that BERT and LogBERT assume; recovery at a masked position t uses only left context plus the mask token at t itself, so the objective is a left-context masked reconstruction variant rather than bidirectional cloze.*

State-space models and Mamba on edge / behavioral telemetry. Gu et al. [5] introduced the structured Normal-Plus-Low-Rank parameterization that reduced state-space model kernel computation to near-linear time, establishing SSMs as a competitive alternative to attention at long sequence lengths. Gu and Dao [4] extended SSMs with input-dependent (selective) dynamics and a hardware-aware parallel scan in Mamba, achieving Transformer-quality performance with linear-time inference. Dao and Gu [22] consolidated the State Space Duality framework in Mamba-2, expressing SSM training as structured matrix multiplication and recovering 2–8× speedup. Kim et al. [6] demonstrate end-to-end edge deployment of Mamba via INT8 quantization and a NAS-optimized accelerator on FPGA / 22nm-ASIC, achieving 4.95–$5.62\times$ lower latency and $48.6\times$ lower energy than baseline accelerators. Xu et al. [7] show that a trained Mamba runs on resource-constrained microcontrollers via a runtime-free C inference engine with operator fusion, with 83% peak-memory reduction. Kong et al. [23] apply a Mamba encoder to insider-threat detection over user-behavior logs with gated fusion and adaptive thresholding. *The closest prior work to this paper is MambaITD; the differences — telemetry modality (kernel eBPF vs. user logs), learning regime (unsupervised next-event NLL vs. supervised classification), and ATT&CK-grounded evaluation (vs. CERT-r5.2 insider-threat labels) — preclude direct numerical comparison.*

Per-event baselines. Liu et al. [24] introduced Isolation Forest, a tree-ensemble anomaly detector that scores points by the expected number of random axis-aligned splits needed to isolate them. We use Isolation Forest as the stateless per-event reference (R2 in §IV-D); the sequence model is meaningful only insofar as it improves over this floor on identical features. *Isolation Forest establishes the per-event signal floor; it cannot exploit temporal context, and the margin between sequence-model and Isolation Forest is the contribution of the sequence model to the same feature representation.*

Positioning. DENDROASPIS connects the sequence-modeling tradition of DeepLog and LogBERT with the linear-time selective-SSM architecture of Mamba, and applies it to structured Tetragon eBPF telemetry. This setting differs from classical syscall-anomaly work, which models only closed syscall-ID sequences, and from log-key anomaly detection, which has not been published on this richer kernel-telemetry representation.

Our contribution is fourfold. First, we show that autoregressive next-event scoring with Mamba produces a strong positive result on kernel telemetry, exceeding chance and unsupervised classical baselines. Second, we document a negative result for vanilla continuous-target masked event modeling: raw detached MSE reconstruction in embedding space collapses on these features. This is a failure mode that masked-modeling work often mitigates with EMA-target mechanisms [21], but that has not been characterized on host-telemetry data. We then show that a field-aware masked objective, with per-field categorical and binary heads, restores performance to the level of next-event NLL. Third, we evaluate against both stateless and stateful baselines, including Isolation Forest, n-grams, supervised XGBoost-LOOCV, and a degenerate XGBoost split that exposes an archive-induced generalization challenge. This lets us measure the value of sequence modeling on identical input features. Fourth, we ground the threat model in an agent-orchestrated, reconnaissance-aware ATT&CK campaign rather than a purely manual red-team setup, with explicit acknowledgment that the agents role is documented as reconstructed post-hoc, not captured from live transcripts.

III. METHODOLOGY

This section specifies the dendroaspis pipeline end-to-end: the input–output contract for unsupervised host-telemetry anomaly detection (§III-A); the two-stage Tetragon-native parser and behavior builder that produce model-ready features (§III-C); the per-field embedding encoder that consumes them (§III-D); the Mamba body that contextualizes them (§III-E); the three detection heads — next-event NLL, vanilla masked event modeling, and field-aware masked event modeling (§III-F); the agent-orchestrated attack campaign and threat model that the detector is evaluated against (§III-G); and the training and inference procedure that constitutes the reproducibility contract (§III-H).

A. Problem formulation

Let $\mathbf{E} = (e_1, e_2, \ldots, e_T)$ be a chronological sequence of kernel-level events emitted by a single host. Each event $e_t \in \mathcal{X}$ is a heterogeneous record drawn from the Tetragon eBPF event schema; $\mathcal{X}$ is the product space of 33 categorical and 9 scalar feature columns (§III-C) plus additional metadata which the model does not consume. A detector assigns a per-event anomaly score $s_t = f_\theta(e_1, \ldots, e_t) \in \mathbb{R}$ such that s_t is large when e_t belongs to an attack and small otherwise; we

evaluate ranking quality directly via AUROC (§IV) and do not threshold during scoring.

We additionally aggregate per-event scores to per-window scores. A window $W_t = (e_{t-L+1}, \ldots, e_t)$ of length $L = 128$ events receives the per-window score $\bar{s}_t = \mathrm{agg}\big(s_{t-L+1}, \ldots, s_t\big)$. Reported as an ablation over $\{\max, \mathrm{mean}, \mathrm{median}, p_{90}, p_{95}, \text{top-}k\text{-mean}\}$ aggregators in §V-C and identify median as the operationally appropriate default (§VI-A). The windowing policy and stride are specified in §III-H.

The detector is trained from *benign* event sequences only; the Atomic Red Team attack intervals [25], [26] are reserved for evaluation (§V).

B. System overview

Figure 1 shows the five-stage pipeline. Raw Tetragon JSON-Lines [3] flow through a Tetragon-native parser (§III-C); a behavior builder then resolves global process lineage and emits 33 categorical and 9 scalar model-input feature columns (§III-C); a per-field embedding encoder (§III-D) maps each event to $\mathbb{R}^{128}$; a 2-layer Mamba body [4] (§III-E) contextualizes the per-window sequence; and one of three heads — next-event NLL, vanilla masked event modeling (MEM), or field-aware MEM — (§III-F) produces the per-event score. The boxes in Figure 1 are labeled with the subsection that specifies each component.

C. Two-stage telemetry pipeline

Stage 1: Tetragon-native parser. The parser ingests gzipped Tetragon JSON-Lines and writes a raw normalized Arrow/Parquet artifact with 50+ kernel-event fields preserved verbatim. Synthetic `process_exec` events that Tetragon emits at agent startup are tagged (rather than dropped) with a boolean flag that the encoder consumes as the model feature `f_is_procfs_walk`. The raw parquet is a persistent artifact, not an intermediate, so downstream feature engineering can be revised without re-parsing ∼19,000,000 JSON-Lines events.

Stage 2: behavior builder. The behavior builder reads the raw parquet in two passes. The first pass streams only $(\texttt{exec_id}, \texttt{parent_exec_id}, \texttt{binary})$ to construct a global process-lineage map; this resolves parent–child chains across files, which is impossible to do per-file because Tetragon does not guarantee a process's `process_exec` record precedes its first `process_kprobe` record in the same file. The second pass derives, for each event, a 14-class action-family abstraction (`PROC_EXEC`, `NET_CONNECT`, `FILE_CHMOD`, `MEM_PROTECT`, ...; cardinality 15 including out-of-vocabulary), bucketed path/IP/object/destination-port categories, lineage features (depth, root-ancestor binary basename hash, parent–child pair hash, process-tree id hash), and log-bucketed timing features (per-process Δt since the same process's previous event, process age). The output parquet has 33 categorical + 9 scalar model-input feature columns (42 channels in total) plus side columns retained for analysis but not consumed by the model.

Pipeline design choices. Five aspects of the pipeline are load-bearing for the model's per-event signal and worth specifying explicitly. (i) The Δt feature is computed per-process rather than against the global event stream, because Tetragon delivers events per-CPU and per-process and a global-stream Δt is dominated by inter-process scheduling noise. (ii) Rows are chronologically sorted within each parquet partition, so the encoder's per-event ordering matches the original kernel-event ordering. (iii) NLL targets are front-pad-aligned: the model's prediction at position t is supervised against the action-family target at t, with the first position masked and the rest of the sequence shifted accordingly. (iv) Lineage is seeded per train/test partition, so within-partition lineage walks are complete; the residual `lineage_missing_parent` rate is 0.44% on train and 2.88% on test, accounted for by processes whose parent predates the partition window. (v) ProcFS-walk events (Tetragon's bootstrap synthetic events from process-tree enumeration at startup) are tagged with a feature flag rather than dropped, so the encoder can decide what to do with them. (vi) Before window construction, the parquet partition is read as a single deterministic stream sorted by $(\texttt{event_time}, \text{source_file_index}, \text{row_index})$, so any two events with identical timestamps still have a stable total order across runs.

Why two stages. Lineage features require global state; emitting them in the parser would tie parser throughput to the speed of the slowest lineage walk. Splitting along this seam keeps Stage 1 streaming and parallelizable, makes Stage 2 idempotent on the persisted raw parquet, and supports encoder iteration without re-parsing.

Label alignment. Each event receives an attack label by joining its parsed Tetragon `event_time` against the labeled-interval table. Interval membership is half-open: $\texttt{start_ts} \le \texttt{event_time} < \texttt{end_ts}$; that is, start inclusive and end exclusive. Events outside all labeled intervals are treated as benign for evaluation. If an event falls inside multiple labeled intervals, `is_attack` remains true and technique attribution is assigned to the matching interval with the most recent `start_ts`; earlier overlapping intervals are shadowed only for technique attribution, not for binary attack labeling. Trial-end cleanup events are labeled as attack when they fall inside the trial interval, because the campaign YAML brackets the full trial lifecycle (reconnaissance → payload → cleanup) under one start/end pair.

D. Per-field embedding encoder

Concretely, the encoder $\phi : \mathcal{X} \to \mathbb{R}^{128}$ is parameterized by 33 embedding tables $\{E_k \in \mathbb{R}^{|\mathcal{V}_k| \times d_k}\}_{k=1}^{33}$, one per categorical or hashed feature, plus 9 scalar pass-throughs for booleans and small numerics (42 model-input channels in total). For an event e_t with categorical values $(v_{t,1}, \ldots, v_{t,33})$ and scalar values $(u_{t,1}, \ldots, u_{t,9})$, the per-field outputs are concatenated to $\mathbf{c}_t \in \mathbb{R}^{277}$ (rich; $\mathbb{R}^{149}$ flat), then projected by a single linear layer $W \in \mathbb{R}^{128 \times 277}$ (rich; $\mathbb{R}^{128 \times 149}$ flat):

$$\mathbf{h}_t = \phi(e_t) = W \cdot \mathrm{concat}\big(E_1[v_{t,1}], \ldots, E_{33}[v_{t,33}], u_{t,1}, \ldots, u_{t,9}\big). \tag{1}$$

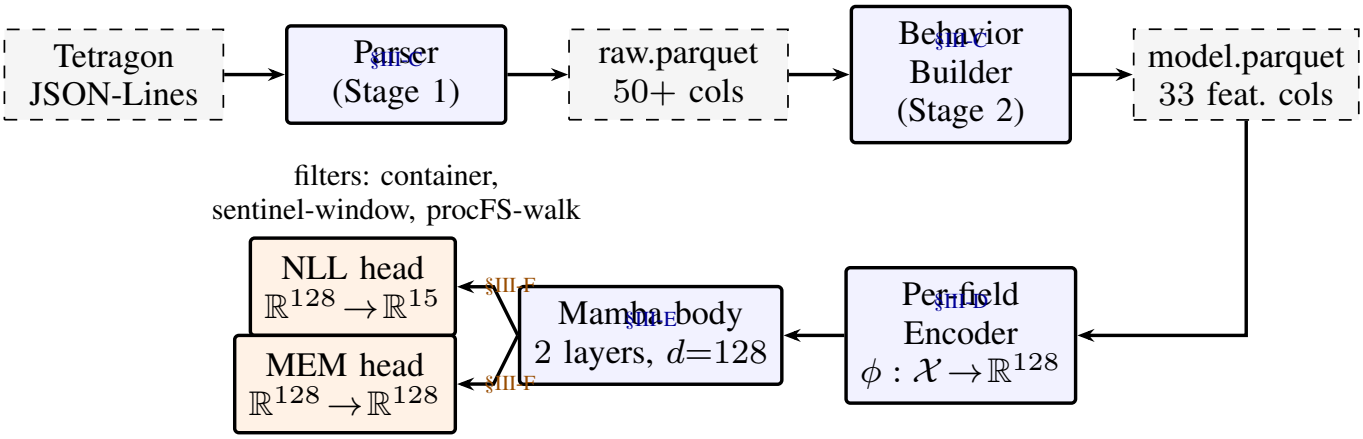

Fig. 1. The pipeline. Tetragon JSON-Lines → event parser (§III-C) → raw normalized parquet → behavior builder (global lineage walk, 14-class action-family abstraction, path/IP/object categorization; §III-C) → model-ready parquet (33 categorical + 9 scalar feature columns (42 total)) → per-field embedding encoder (§III-D) → 2-layer Mamba body (§III-E) → one of three heads: field-aware MEM (primary), vanilla MEM (control), or NLL (§III-F). Persisted parquet artifacts (between stages) make every downstream stage independently re-runnable.

Each E_k has a dedicated index reserved for null/out-of-vocabulary values, so missing fields are encoded as a learned vector rather than silently mapped to the modal class. The two highest-dimensional embedding tables are proc-name (2048 buckets, 32-dim) and parent-proc-name (1024 buckets, 32-dim) hashes; 19 small categorical and bucketed-scalar tables share a common 4-dim embedding (cardinalities 3–16). The full cardinality and dimension table is in the released config.

Per-field, not per-joint-token. Each categorical or hashed feature has its own embedding table; the projection and body learn cross-field interactions. This avoids combinatorial sparsity in any single hashed joint-token vocabulary.

Feature-set parameter (`rich`, `flat`). The encoder accepts a `feature_set` kwarg that selects between two column inventories: `rich` (default, 42 channels — 33 categorical + 9 scalar) and `flat` (35 channels — 26 categorical + 9 scalar, dropping seven high-cardinality identity hashes: `f_proc_name_hash`, `f_parent_proc_hash`, `f_proc_cwd_hash`, `f_lineage_bag_hash`, `f_parent_child_pair_hash`, `f_root_ancestor_basename_hash`, `f_process_tree_id_hash`). The detection heads (§III-F) are unchanged across feature sets; only the encoder's input column list and the projection layer's input dimension differ (277 → 149 pre-projection concat dim under flat; encoder parameter count 262,052 → 20,388).

E. Mamba body

Standard. We use a 2-layer pre-norm Mamba state-space block [4], [5] with model dimension $d = 128$, applied directly over the per-event embedding sequence $\mathbf{H} = (\mathbf{h}_1, \ldots, \mathbf{h}_L)$ produced by the encoder. The block follows the residual form $\mathbf{x} \leftarrow \mathbf{x} + \mathrm{block}(\mathrm{LN}(\mathbf{x}))$ and is followed by a final LayerNorm and dropout with $p = 0.1$. We use the `mamba-ssm` selective scan implementation with causal-conv1d; dependency pins are reported for reproducibility.

TABLE I
THREE SCORER VARIANTS ATTACHED TO THE SHARED ENCODER AND MAMBA BODY. EACH IS SELECTABLE VIA A SINGLE TRAINING-SCRIPT FLAG (`--OBJECTIVE {MEM-FA, MEM, NLL}`). THE BODY, MASKING STRATEGY (WHEN APPLICABLE), AND CHUNKED ALL-POSITIONS EVALUATION PROTOCOL ARE SHARED; ONLY THE HEAD AND LOSS DIFFER.

Obj.	Head	Loss
`nll`	Linear$(d, 15)$, predicts next-event `f_action_family`	Cross-entropy at each position vs. shifted target
`mem`	Linear(d, d), predicts pre-mask encoder embedding at masked positions	Continuous MSE against detached encoder target
`mem-fa`	5 CE heads (`action_family`, `kprobe_function`, `path_category`, `dst_port_category`, `object_category`) + 4 BCE heads (`uid_eq_parent`, `kp_commit_creds_uid_change`, `kp_commit_creds_cap_change`, `is_setuid_exec`) at masked positions	Uniform mean across 9 per-field losses

F. Detection heads

The body's contextualized output $\mathbf{Z} = (\mathbf{z}_1, \ldots, \mathbf{z}_L) \in \mathbb{R}^{L \times 128}$ feeds one of three heads. We train them as separate models that share encoder and body architectures but not weights, so any difference at evaluation time isolates the choice of objective. Table I summarizes the three variants.

NLL head. A linear classifier $\mathbb{R}^{128} \to \mathbb{R}^{15}$ predicts the action-family of the next event from the contextualized representation of the current event:

$$\mathcal{L}_{\mathrm{NLL}} = -\frac{1}{B(L-1)} \sum_{b,t} \log p_\theta\big(y_{b,t+1} \,\big|\, \mathbf{z}_{b,t}\big)\,, \qquad (2)$$

where $y_{b,t+1} \in \{1, \ldots, 15\}$ is the action-family target. The per-event anomaly score is the per-position cross-entropy at evaluation time, $s_t = -\log p_\theta(y_{t+1} \mid \mathbf{z}_t)$. This is the DeepLog [2] unsupervised next-event paradigm, with action-family as target rather than a hashed joint token.Following standard autoregressive convention, NLL is computed over predictions at positions $1, \ldots, L-1$ within each length-L window (logits at position t predict the event at $t+1$); position 0 is front-padded with a placeholder score so that per-event indices align with the original event stream when joining scores to evaluation labels.

MEM head. Vanilla continuous-target masked event modeling. At training time, 15% of the positions in each window are selected uniformly at random; the encoder embedding $\mathbf{h}_t$ at each masked position is replaced with a learned $\mathbb{R}^{128}$ `[MASK]` parameter before entering the body. A linear head $\mathbb{R}^{128} \to \mathbb{R}^{128}$ predicts the original (pre-mask) embedding at masked positions; the loss is mean squared error against the *detached* encoder target:

$$\mathcal{L}_{\text{MEM}} = \frac{1}{|\mathcal{M}|} \sum_{(b,t)\in\mathcal{M}} \left\| \mathbf{p}_{b,t} - \text{sg}(\mathbf{h}_{b,t}) \right\|_2^2, \tag{3}$$

where $\mathcal{M}$ is the set of masked positions, $\mathbf{p}_{b,t}$ is the head's prediction, and $\text{sg}(\cdot)$ is the stop-gradient operator. The masking and continuous-target reconstruction draw on data2vec [21], but a key distinction from BERT-style cloze [19] is that the Mamba body is causal: reconstruction at a masked position t uses only left context (positions $< t$) plus the mask token at t itself, not the bidirectional context that BERT and LogBERT assume, which is deferred to future work. The objective is therefore a left-context masked reconstruction variant, not bidirectional cloze.

Field-aware MEM head - MEM-FA (primary). The continuous MSE objective above has three failure modes on this feature set: high-norm hash dimensions dominate the loss, sparse security-relevant fields are averaged out, and the encoder is free to collapse all events toward a low-norm region of $\mathbb{R}^{128}$ without paying loss. The field-aware variant replaces the single regression head with nine per-field heads — five categorical cross-entropy heads on (`f_action_family`, `f_kprobe_function`, `f_path_category`, `f_dst_port_category`, `f_object_category`) and four binary cross-entropy heads on (`f_uid_eq_parent`, `f_kp_commit_creds_uid_change`, `f_kp_commit_creds_cap_change`, `f_is_setuid_exec`). The loss is the uniform mean of the nine per-field losses; the per-event anomaly score at evaluation time is the same uniform mean. Per-field heads recover sparse security markers (capability changes, setuid execs) that vanilla MSE diluted, and the categorical-target structure forces the encoder to represent each field's identity rather than collapse it.

Why these nine fields. The five categorical heads cover the high-frequency behavioral context fields a host-anomaly model must represent: action family, kprobed kernel function, path category, destination-port category, object category. The four binary heads cover sparse but security-relevant markers — parent–child UID equality, two credential-mutation flags, and setuid execs — that vanilla MSE dilutes because their per-event MSE contribution is dwarfed by the 128-d embedding average. The seven high-cardinality identity hashes (`f_proc_name_hash`, ...) are retained as encoder *inputs* but excluded as reconstruction *targets*: predicting a 1024-bucket hash from local context degrades to memorization rather than learning behavioral structure.

Eval-time interleaved-mask scoring. Both MEM variants need a deterministic test-time protocol because random training-time masking would leave 85% of positions with no reconstruction error. We use a chunked all-positions scoring scheme: for each window of length $L = 128$ and chunk size $n_{\text{chunks}} = 8$, the model is run n_{chunks} times with disjoint masks $m_c = \{t : t \bmod n_{\text{chunks}} = c\}$ for $c \in \{0, \ldots, 7\}$. For a causal Mamba body, contiguous masks deny each masked position any local context whereas interleaved masks preserve a same-distribution of the preceding local context, matching the per-event noise floor seen at training time. After the n_{chunks} passes every position has been masked exactly once, the per-pass mask density is $1/n_{\text{chunks}} = 12.5\%$ (close to the training-time 15%), and per-event score is the per-position field-loss (or MSE) at the unique pass that masked it.

G. Agent-Orchestrated Attack Campaign

The attack workload was constructed as an ATT&CK-aligned campaign executed in a controlled Proxmox/LXC developer environment. The campaign used LLM-assisted operational planning to select and sequence Linux-relevant offensive primitives drawn from the Atomic Red Team catalog [25], including techniques conditioned on the observed target environment — for example, the secondary systemd-service technique `T1543.002-2` was dropped after the planning step detected a Debian distro mismatch in the LXC instance, whereas the primary `T1543.002` variant was retained. The campaign yaml documenting technique selection is released alongside the corpus.

We characterize the agent's role as *type-1 adaptation*: pre-campaign environmental tailoring with reconnaissance-aware technique selection, distinct from *type-2* within-campaign feedback-loop adaptation. The agent observed the target environment and selected technique combinations appropriate to it; the agent did not observe trial outcomes within the campaign and adapt subsequent trial selection. The pacing of execution is consistent with machine-mediated rather than human-mediated dispatch: the inter-trial gap distribution measured directly from the labeled-interval timestamps has a median of 11.9 seconds with 80% of consecutive trials spaced under 60 seconds (§V-F, Fig. 5); a manually-operated red-team campaign typically shows median gaps in the 5-minute range.

Two specific scenarios in the labeled set. Beyond the standard ATT&CK atomics, the campaign includes two scenarios designed to anchor the threat-model framing. (i) Track A is a synthetic npm-postinstall supply-chain compromise: a research-artifact npm package whose `postinstall` hook

spawns an `npm` → `node` → `sh` → `curl` process lineage, reads synthetic credentials from a sandbox directory (`~/.fake_secrets/`), base64-encodes them and POSTs them to a loopback listener at `127.0.0.1:8888`, and drops a cron persistence entry that is cleaned up at trial end (8 labeled trials, surfaced under `T1195.002`). The artifact is a research stand-in for the npm-postinstall compromise class — the same class as the publicly-disclosed Shai-Hulud worm [9], though without the latter's worm propagation or real-internet exfil. No real credentials, paths, or external endpoints are touched. (ii) `CHAIN.CAP01` is a chained post-compromise scenario combining five Atomic Red Team primitives in a single labeled interval: shell spawn, process discovery, credential read, network ingress, and evidence cleanup (2 labeled trials).

Because the original LLM prompts and outputs were not fully serialized at execution time, the agentic role is reported as a reconstructed methodology rather than a fully reproducible closed-loop autonomous trace. The reconstruction is supported by (a) the campaign yaml documenting technique selection, (b) the labeled trial structure and the distro-gating evidence, and (c) the inter-trial cadence in the labeled timeline. A future collection will rerun a representative subset under full prompt/output logging with a human-paced control condition (§VII-A).

Threat model.

- *Defender:* Mamba detector trained on benign developer telemetry; one per host.
- *Attacker:* LLM-assisted operational planning using ATT&CK-aligned offensive primitives from the Atomic Red Team catalog [25]; reconnaissance-aware technique selection but no within-campaign feedback adaptation.
- *Environment:* Proxmox/LXC developer workstation running representative IDE, language-server, package-manager, shell, and container-build workloads.
- *Telemetry:* Tetragon eBPF event stream [3], in-kernel filtering followed by user-space normalization (§III-C).
- *Goal:* Determine whether per-event sequence modeling separates attack-window events from noisy developer-baseline events under this threat model.

H. Training and inference

The reproducibility contract for all three heads:

Optimization. AdamW [27] with $\mathrm{lr} = 3 \times 10^{-4}$, weight decay 0.01, $\beta = (0.9, 0.95)$, gradient clipping at 1.0. The learning rate uses a cosine schedule with 10% linear warmup over the first 10% of total training steps. Mixed precision uses bfloat16 autocast on the GPU. Training runs for at most 15 epochs with early stopping (patience 5 epochs, $\min_\mathrm{delta} = 10^{-4}$ on validation loss).

Windowing and batching. Sliding $L = 128$ event windows with stride 32 (75% overlap), so each event appears in up to 4 training windows. Per-epoch shuffling is applied at the window level. Train batch size 256 with 6 DataLoader workers and prefetch factor 4; evaluation batch size 512 with the same window length.

Validation carve. The training partition is split chronologically into 80% train and 20% validation, with a 15-minute hold-out gap straddling the split timestamp; events in the gap are dropped from both partitions to prevent any training window and any validation window from overlapping in time. The test partition is held out at the corpus level (§V).

Leakage controls. No labeled attack interval falls within the training or validation partition; the labeled ATT&CK campaign is contained entirely within the test partition by construction (§V). Attack labels are used only for evaluation and never for model selection — early stopping and hyperparameter choices use the validation-loss signal on benign-only data, not any attack-label signal.

Reproducibility. The corpus is released on Hugging Face Datasets at https://huggingface.co/datasets/rypow/dendroaspis-tetragon-hids [10] (revision `a89608d9...`; CC BY 4.0). Training and eval config and code is released on github at https://github.com/ryypow/dendroaspis [11]; running the released pipeline with the released manifest and the recorded random seeds reproduces the headline numbers in Table II to within bootstrap noise.

Compute. A consumer-grade GPU; peak training memory at batch size 256 stayed below 10 GB. Software: Python 3.12, PyTorch 2.10, `mamba-ssm` 2.3.1 with `causal-conv1d` 1.6.1; full dependency pins in the released repository's `requirements.txt`.

IV. Experimental Design

This section defines the evaluation protocol that produces the experiments in §V and the figures in §V. We describe the corpus and splits (§IV-A), the primary and ablation baselines we compare against (§IV-B), the metrics we report (§IV-C), the experiments and their falsification criteria (§IV-D).

A. Dataset and splits

Class Distribution. The corpus contains 19,301,915 events across 142 archive files. Of those, 16,951,950 events (87.83%) form the training partition and 2,349,965 events (12.17%) form the held-out test partition. The training partition is further chronologically carved into 80% train and 20% validation as described in §III-H. The test partition contains 705,852 attack events labeled by the ATT&CK-aligned, agent-orchestrated campaign described in §III-G, covering 22 ATT&CK techniques — including a synthetic npm-postinstall supply chain compromise (Track A:T1195.002, 8 labeled trials, 15,703 events) modeled after the same threat class as the Shai-Hulud npm worm [9] but without its self-propagating or real exfiltration — plus a chained post-compromise scenario (`CHAIN.CAP01`: shell → discovery → credential read → loopback exfil → cleanup; 34,130 events) in the test split.

Split policy. The train/test split is *archive-based*: archives captured before the labeled-attack window (≤ `2026-04-19T22:09:18Z` last event) form train; archives captured during the frozen labeled-attack window (`2026-04-19T21:46:27Z` onward) form test. All 61

labeled ATT&CK intervals fall inside test by construction, with no positive labels available in the train fold (§IV-B, XGBoost honest framing). Time-based splitting is the appropriate choice for a host-based detection task because random splitting would leak future process-state and lineage information into past windows [28].

Label source. Labels are derived from 61 ATT&CK trial intervals recorded in `labels.csv`. Each interval is bracketed by a pair of file-creation markers — one written immediately before `Invoke-AtomicTest` fires the technique, one immediately after it returns — whose timestamps the eBPF sensor captures inline with the rest of the event stream. Every event whose `event_time` falls inside a marker-bounded interval is labeled an attack event; all other test-partition events are labeled benign. We do not perform manual per-event review. Because the markers bracket the harness invocation rather than the technique itself, harness scaffolding (PowerShell setup, environment preparation, cleanup) is labeled alongside the malicious action. We name this labeling-granularity issue as a construct-validity threat in §VI-D.

B. Baselines

We report a hierarchy of baselines spanning trivial, classical, supervised-classical, and the two Mamba scorer variants. Each baseline operates on the same time-based test partition and is evaluated under the same metrics (§IV-C). The baselines deliberately differ in how much of the encoder's channel set they consume and whether they preserve sequence context; the framing of each comparison below names the specific axis being isolated rather than treating the baselines as undifferentiated alternatives.

Channel-and-sequence framing. The encoder produces 42 model-input feature columns (33 embedded categorical/hashed columns plus 9 float pass-throughs; §III-D). Each baseline below reads a strict subset of these channels and, separately, either preserves or drops the per-event sequence ordering. The Mamba detector consumes all 42 channels with full sequence context; the baselines below consume one or the other but not both. We make the channel-coverage / sequence-coverage trade explicit so each Mamba-vs-baseline margin isolates a specific axis of the contribution rather than confounding two.

Trivial floor: random ranking. A uniform random per-event score has expected $\mathrm{AUROC} = 0.5$ and $\mathrm{AP} \approx 0.30$ at this corpus's positive rate — 30% `of the test set consisting of events from within marked attack intervals`. We report the analytic value as the chance line.

Classical unsupervised: n-gram on the joint event token. A trigram language model on the joint `token` column with Laplace smoothing ($\alpha = 1.0$) and stupid backoff [29] (backoff factor 0.4), trained on the train-fold benign sequence and used at test time as a sequence-NLL scorer. Sequence boundaries are per-process (`proc_exec_id`). The `token` column is built from 6 of the 42 encoder channels — `action_family`, `proc_basename`, `parent_basename` (only on `PROC_EXEC`), `path_category` (only on `FILE_*`), `dst_ip_category` and `dst_port_category` (only on `NET_*`) — collapsed into a single per-event string. The n-gram therefore preserves sequence context but drops 36 of the 42 channels Mamba sees. At scoring time, an event with insufficient training context (unseen (w_{t-2}, w_{t-1}) prefix or beginning-of-process) backs off to the bigram or unigram estimate per the citation above. *Any margin Mamba achieves over the n-gram is therefore the combined contribution of (a) channel access and (b) sequence modeling; the comparison is not an isolated test of the selective state-space body.*

Classical unsupervised: Isolation Forest . An Isolation Forest [24] on the same 42 encoder feature columns the Mamba scorers consume, but as a flat per-event matrix with no sequence context. Fit on the full benign training partition (16,951,950 events) with `n_estimators` $= 200$, `max_samples` $=$ auto ($\min(256, n)$ per scikit-learn [30]), `contamination` $=$ auto, `random_state` $= 0$. *Any margin Mamba achieves over Isolation Forest is therefore the isolated contribution of the sequence model on the same 42-channel representation.* This is the cleanest single-axis baseline.

Supervised-classical: XGBoost on window aggregates. A gradient-boosted decision-tree classifier [31] trained on per-window aggregates of the same 42 feature columns against the per-window attack label. We include this baseline to characterize the supervised-baseline failure mode under the train/test split: because the labeled ART campaign falls entirely in the test fold by construction (§IV-A), the XGBoost `fit` call hits its degenerate-mode branch (no positive windows available in train), and at score time the classifier returns zeros. We report the time-split XGBoost result only as a negative control for label availability, not as a competitive supervised baseline.

Mamba MEM-FA. The primary detector under the field-aware MEM objective (§III-F), trained on the same corpus with the same encoder. Same all-42-channels-plus-selective-sequence input as Mamba NLL. Single seed.

Mamba NLL. The primary detector under the next-event NLL objective (§III-F), trained on the corpus with the front-pad alignment and the same encoder. Consumes all 42 channels with full sequence context. Single seed.

What the four-baseline pattern lets us measure. The Mamba-vs-IF margin isolates the sequence-model contribution on the same 42-channel representation. The Mamba-vs-n-gram margin combines channel access and sequence modeling. The IF-vs-n-gram margin (which is itself a comparison between two unsupervised baselines on the same corpus) approximates the trade between "all 42 channels but no sequence" and "6 channels but with sequence" — a useful sanity check on which axis carries more per-event signal at the classical-baseline level. The XGBoost row sits outside the unsupervised lineup as a label-availability marker.

Encoder $\times$ architecture decomposition triplet. Three additional baselines probe the encoder $\times$ architecture decomposition reported in §V-D. Each is a re-run of one of the

existing baseline classes under a controlled axis change rather than an additional method choice.

Isolation Forest (flat encoder). The same Isolation Forest [24] as above, run on the 35-channel flat-encoder feature matrix (the seven high-cardinality identity hashes dropped per §III-D). Identical hyperparameters and training-sample protocol to IF-rich. Pairs with Mamba-flat to isolate the contribution of the seven hash columns to each model class.

Mamba MEM-FA (flat encoder). A retrained MEM-FA Mamba on the 35-channel flat-encoder representation. Identical hyperparameters and training protocol to MEM-FA-rich (§III-H); only the encoder `feature_set` is changed. Single seed.

XGBoost (LOOCV supervised reference). A leave-one-trial-out cross-validated XGBoost classifier on per-event 42-channel rich features. For each of the 61 labeled attack intervals, train one classifier on (benign-train events + events from the 60 other labeled trials, the latter as positive labels) and score the held-out trial; stitch the 61 held-out per-event score arrays into a single per-event score array. This baseline is a generous *upper bound* rather than a peer: it sees per-event attack labels at training time and benefits from a documented technique-level information leak (sibling trials of the held-out technique remain in training). The leak inflates per-event AUROC to 0.8041 and best-aggregator per-window AUROC to 0.8421 (mean aggregator), against 0.6234 at the max aggregator. We retain it for the matched-aggregator comparison in §V-C, where it serves as the supervised-classification ceiling.

Why no published-SOTA host-telemetry baseline. We do not include LogBERT [20], NeuralLog [16], or DeepLog [2] re-runs. The published implementations target a different modality (HDFS / BGL / OpenStack log keys at ~ 300–1000 vocabulary), have no released host-telemetry preprocessing pipeline, and re-implementing them on the dendroaspis features would require committing to a tokenization scheme that is itself an open research question (§VI-E). We discuss this paradigm gap in §II and frame it as future work rather than present a contrived re-run.

C. Metrics

We report four metrics, each chosen for a specific role.

Per-event AUROC (primary). The area under the receiver operating characteristic curve over per-event scores. AUROC has a clean sampling interpretation as the probability that a uniformly random attack event scores higher than a uniformly random benign event, and is invariant to monotonic score transformations. With 2,349,965 test events, AUROC has very tight bootstrap intervals; we report 95% bootstrap confidence intervals with 1000 resamples.

Average precision (AP, secondary). The area under the precision-recall curve. AP is reported because AUROC can mask poor precision in the high-recall regime relevant to operational alerting. The random-classifier AP baseline is the positive rate, $\mathrm{AP}_{\mathrm{rand}} \approx 0.30$ at this corpus's class balance.

TPR at 1% FPR (operational). The fraction of attack events ranked above the threshold that catches 1% of benign events. At 1% FPR on 1,644,113 benign test events, the alert volume is $\sim$16,441 false positives over the test window — the upper end of what a single SOC analyst could realistically triage.

Per-window AUROC (secondary, diagnostic). Per-window scores are produced by aggregating per-event scores within each window (L=128, S=32). We report per-window AUROC under both the $\max$ aggregator (uniform-default for cross-model comparison) and each model's best-aggregator (matched comparison; §V-C); the matched-aggregator value is the headline for the Mamba-vs-classical claim. The full aggregator set $\{\max, \mathrm{mean}, \mathrm{median}, p_{90}, p_{95}, \text{top-5-mean}, \text{top-10-mean}\}$ is exercised by a ablation (§IV-D, E3) that quantifies how much of the per-window signal is aggregator-determined vs. model-determined.

Per-technique AUROC. For each of the 22 labeled ATT&CK categories we compute the AUROC over the union of (i) that technique's attack events and (ii) all benign test events. Per-technique AUROC identifies which attack classes the detector resolves cleanly and which it cannot; it is also the level at which the labeling-granularity threat surfaces (§VI-D).

D. Experimental protocol

We define five experiments. Each has a question, a procedure, and a falsification criterion that determines whether the experiment counts as a positive or negative result independent of the eventual numbers.

E1: Does NLL Mamba detect ATT&CK attacks? Train the NLL head on the training partition with the encoder of §III-D; evaluate on the test partition. Falsification criterion: per-event AUROC < 0.60 — below this threshold the parser-and-feature pipeline would not have produced a usable per-event signal, and we would attribute the result to objective-level rather than corpus-level limits.

E2: Does field-aware MEM detect ATT&CK attacks? Train the MEM-FA head (§III-F) on the same training partition; evaluate on the same test partition with the deterministic interleaved-mask all-positions protocol (§III-F). Falsification criterion: per-event AUROC < 0.60 — below this, the field-aware target would not have produced per-event signal beyond the vanilla MEM baseline reported in §V-A, and the headline parity claim against NLL would be unsupported.

E3: Does the choice of per-window aggregator change the headline per-window AUROC? Compute per-window AUROC under each aggregator $\mathrm{agg} \in \{\max, \mathrm{mean}, \mathrm{median}, p_{90}, p_{95}, \text{top-5-mean}, \text{top-10-mean}\}$ for both NLL and MEM-FA scorers on the same per-event score arrays. Pre-registered question: does the $\max$ aggregator under-estimate per-window AUROC by more than 0.10 relative to the best alternative, for either scorer? An affirmative answer would imply that any per-window inversion under $\max$ is aggregator-bounded rather than scorer-bounded.

R2: Does the sequence model add value over the per-event Isolation Forest baseline? Compare each Mamba scorer's per-event AUROC against the Isolation Forest baseline

on identical features. *Falsification criterion:* $\mathrm{AUROC}_{\mathrm{mamba}} - \mathrm{AUROC}_{\mathrm{IF}} < 0.05$ for both scorers — below this, the sequence model is not adding meaningful per-event signal over a stateless tree ensemble on the same features.

R3 (post-hoc): Does dropping the seven high-cardinality identity-hash columns from the encoder change Mamba's per-event AUROC? A post-hoc encoder ablation motivated by the IF-flat result (§V-D): the matched-encoder Isolation Forest run shows the seven hash channels are net noise to per-event density estimation, and the question is whether the same holds for the Mamba sequence body. Train MEM-FA Mamba under the flat encoder (§III-D) with otherwise-identical hyperparameters; evaluate on the same test partition under the same protocol. Falsification criterion: if Mamba-flat per-event AUROC is more than 0.02 below Mamba-rich, hash channels are load-bearing for Mamba and the encoder-by-architecture decomposition does not extend across model classes. We report the result regardless because it inverts the rich-vs-flat expectation; the post-hoc framing is named in §VI-E alongside the single-seed protocol.

Ablation policy. We do not run a per-feature ablation in this paper. The encoder concatenates 33 categorical embeddings plus 9 scalar pass-throughs (§III-D); a single-feature ablation set would require re-training 42 models, and a single-seed protocol would not produce conclusive comparisons. The omission is named in §VI-E as a known limitation; per-technique AUROC (§IV-C) and the channel-coverage delta between the four baselines (§IV-B) are the qualitative substitutes.

V. Results

This section reports the five pre-registered experiments of §IV-D on the released corpus. §V-A reports E1 (NLL Mamba), E2 (MEM-FA Mamba), and the classical-baseline floor against which both scorers are measured. §V-B disaggregates by ATT&CK technique (companion figure in §V-B). §V-C reports the per-window aggregator ablation (E3). §V-D reports the encoder $\times$ architecture decomposition (R3). §V-E reports the per-event score-distribution structure that produces the headline numbers. §V-F reports the agent-cadence measurement that anchors the threat-model narrative of §III-G. §V-G collects the honest negatives that sit alongside the positive findings. R2 (sequence-model lift over Isolation Forest) is addressed inline in §V-A. The headline numbers are summarized in Table II; per-technique numbers are in Table III and Figure 2; the aggregator ablation is in Table IV; the cadence histogram is Figure 5.

A. Headline detection results (E1, E2, R2)

E1: NLL Mamba clears its falsification threshold. The NLL Mamba reaches per-event $\mathrm{AUROC} = 0.741$ (95% CI $[0.740, 0.742]$), $\mathrm{AP} = 0.619$, and $\mathrm{TPR@1\%FPR} = 0.159$ over the 2,349,965-event test partition. The pre-registered E1 criterion (§IV-D) was per-event $\mathrm{AUROC} \geq 0.60$; the result clears the threshold by $+0.141$, so E1 reads as a positive result. The operational metric $\mathrm{TPR@1\%FPR}$ at 0.159 is the load-bearing reading of this experiment for triage-volume budgeting: $\sim$16,441 false positives at the 1% FPR threshold against $\sim$112,230 true positives over the test window. The front-pad alignment of the NLL target against the encoded current-event features (§III-F) is the design decision the operating-point number is most sensitive to.

E2: MEM-FA Mamba clears its falsification threshold and edges past NLL on AUROC. The MEM-FA Mamba reaches per-event $\mathrm{AUROC} = 0.751$ (95% CI $[0.751, 0.752]$), $\mathrm{AP} = 0.488$, and $\mathrm{TPR@1\%FPR} = 0.021$. The pre-registered E2 criterion was per-event $\mathrm{AUROC} \geq 0.60$; MEM-FA clears it by $+0.151$, so E2 reads as a positive result. Headline AUROC narrowly exceeds NLL by $+0.010$, but the two scorers diverge sharply on operational metrics (NLL's AP is $+0.131$ higher and its $\mathrm{TPR@1\%FPR}$ is $\sim 8\times$ higher). The two-objective-divergence is a finding in itself: NLL produces a more concentrated top-of-list, MEM-FA a flatter score distribution, and the two are mechanistically distinct paths to a common global ranking score (§V-B, §VI-A). Vanilla per-event MEM (single per-event reconstruction-error head over the concatenated encoder vector, no per-field decomposition) was trained under the same protocol as a control and collapsed to a degenerate solution — per-event reconstruction error became approximately uniform across attack and benign events, $\mathrm{AUROC} \approx 0.5$ — because a single regression head over a 277-dimensional concatenation of categorical embeddings and float pass-throughs has no incentive to spend capacity on the discrete identity fields that carry the per-event anomaly signal. The field-aware redesign (§III-F) targets this collapse mode directly; MEM-FA's 0.751 headline is therefore the masked-modeling baseline that actually trains on this corpus.

R2: Mamba beats the per-event Isolation Forest baseline on identical features. Isolation Forest (rich) reaches $\mathrm{AUROC} = 0.666$ (95% CI $[0.665, 0.666]$). NLL Mamba clears the $+0.05$ lift criterion by $+0.075$; MEM-FA Mamba (rich) clears it by $+0.085$. R2 reads as a positive result for both scorers: the selective state-space body adds meaningful per-event signal over a stateless tree ensemble on the same 42-channel feature representation. At matched flat-encoder (35-channel) representation the gap widens to $+0.133$ (§V-D, R3).

R3: Dropping the seven identity-hash columns from the encoder lifts Mamba MEM-FA above rich. A retrained MEM-FA Mamba on the flat 35-channel encoder (the seven high-cardinality identity hashes dropped per §III-D) reaches per-event $\mathrm{AUROC} = 0.782$ (95% CI $[0.782, 0.783]$). The rich-vs-flat 95% CIs are disjoint ($[0.751, 0.752]$ vs. $[0.782, 0.783]$); the $+0.031$ flat-over-rich lift is not bootstrap noise. The pre-registered 0.02 post-hoc falsification criterion (§IV-D, R3) requires Mamba-flat to land within 0.02 of Mamba-rich; the actual result is $+0.031$ above. Hash channels are net noise to Mamba's sequence body on this corpus, not load-bearing. Full encoder $\times$ architecture decomposition in §V-D.

Classical-baseline floor. The n-gram trigram on the 6-channel joint event token (§IV-B) reaches $\mathrm{AUROC} = 0.567$, materially above chance but ~ 0.21 below the Mamba-flat headline. Isolation Forest (rich) at 0.666 is the stronger of the two unsupervised classical baselines but still ~ 0.116

TABLE II

MAIN DETECTION RESULTS ON THE HELD-OUT TEST PARTITION (2,349,965 EVENTS, 30.04% ATTACK). ALL SCORERS OPERATE ON IDENTICAL INPUT FEATURES WITHIN EACH RICH OR FLAT ENCODER (§III-D). AUROC AND PER-WINDOW AUROC USE THE PER-EVENT SCORE AND THE PER-WINDOW max-AGGREGATED SCORE RESPECTIVELY; PER-WINDOW AUROC UNDER EACH MODEL'S MATCHED (BEST) AGGREGATOR IS REPORTED SEPARATELY IN TABLE IV, AND THE MATCHED-AGGREGATOR HEADLINE FOR THE MAMBA-VS-CLASSICAL CLAIM IS DISCUSSED IN §V-C. 95% CONFIDENCE INTERVALS ARE BOOTSTRAP PERCENTILES OVER 1000 RESAMPLES. AP RANDOM BASELINE IS ≈0.30 AT THIS CORPUS'S POSITIVE RATE. **BOLD** = BEST *unsupervised* (PAPER'S PRIMARY CONTRIBUTION); UNDERLINE = NEXT-BEST UNSUPERVISED ON PER-EVENT AUROC. THE XGBOOST (TRAIN-FOLD DEGENERATE) ROW RECORDS THE SUPERVISED BASELINE FAILURE MODE UNDER THE ARCHIVE-BASED TRAIN/TEST SPLIT: THE LABELED CAMPAIGN FALLS ENTIRELY ON THE TEST SIDE BY CONSTRUCTION (§IV-A), SO NO POSITIVE WINDOWS ARE AVAILABLE AT FIT TIME. THE XGBOOST (LOOCV) ROW IS A GENEROUS SUPERVISED *ceiling reference*, NOT AN UNSUPERVISED PEER (§IV-B). $n_{\text{SEEDS}} = 1$ FOR ALL MAMBA RUNS.

Method (experiment)	Per-event AUROC	AP	TPR@1% FPR	Per-window AUROC (max)
Random ranking (analytic floor)	0.500	0.300	0.010	0.500
n-gram trigram (joint event token)	0.567 [0.566, 0.568]	0.392	0.040	0.401 (inverted)
Isolation Forest (rich, 42-ch, R2 ref.)	0.666 [0.665, 0.666]	0.366	0.004	0.309 (inverted)
Isolation Forest (flat, 35-ch)	0.649 [0.649, 0.650]	0.361	0.004	0.305 (inverted)
XGBoost (supervised, train-fold degenerate)	0.500 [0.500, 0.500]	0.300	0.000	0.500
NLL Mamba (E1, rich)	0.741 [0.740, 0.742]	**0.619**	**0.159**	**0.555**
MEM-FA Mamba (E2, rich)	<u>0.751</u> [0.751, 0.752]	0.488	0.021	0.501
MEM-FA Mamba (R3, flat encoder)	**0.782** [0.782, 0.783]	0.505	0.018	0.528
XGBoost (LOOCV ceiling ref.)*	0.804 [0.803, 0.805]	0.599	0.009	0.623

* Supervised oracle: sees per-event attack labels at training time. Generous ceiling, not an unsupervised peer (§IV-B).

below Mamba-flat; the flat-encoder Isolation Forest at 0.649 regresses slightly below rich, the IF half of the encoder × architecture decomposition (§V-D). XGBoost evaluates at AUROC = 0.500, the degenerate floor of the supervised setting under the archive-based train/test split: with the labeled ART campaign held entirely in test by construction (§IV-A), the XGBoost `fit` call hits its no-positive-windows branch and at score time returns zeros uniformly. We report this honestly; it is a property of the train/test boundary and a legitimate negative for supervised baselines on this corpus design, not a defect of XGBoost as a model class. The leave-one-trial-out XGBoost variant (LOOCV ceiling reference, 0.804) recovers a real supervised number under cross-validation; we discuss it as a generous ceiling rather than a peer (§IV-B, §V-C).

Per-window AUROC under max is at or below chance for every unsupervised model. Every unsupervised entry in the rightmost column of Table II is at or below 0.555, with the two unsupervised classical baselines inverted (n-gram 0.401, IF 0.309). The supervised LOOCV oracle at 0.623 is the only per-window-max entry above 0.555. The headline per-window numbers under max-aggregation appear to leave little room for a sequence-model contribution. §V-C shows that this is mostly an aggregator artifact, not a model-quality artifact.

B. Per-technique structure

Table III reports per-technique AUROC for both Mamba scorers and the two unsupervised classical baselines across all 22 labeled ATT&CK categories, sorted by attack-event count (largest first). Figure 2 visualizes the same data.

Three findings stand out.

(i) NLL inverts on Unix-Shell techniques; MEM-FA recovers (the labeling-granularity finding). On `T1059.004` (Unix Shell — bash), NLL produces AUROC = 0.479, sub-random by 0.021. The technique accounts for 175,797 attack events — 24.9% of all attack events in the test partition — so the inversion materially drags the global AUROC. MEM-FA on the same events recovers to AUROC = 0.616: still the lowest per-technique number under MEM-FA, but no longer below chance. The mechanism we attribute to this finding is *labeling granularity*: ART harness scaffolding (the dispatcher process, the trial-bracketing bash invocations, and the cleanup) is co-labeled with the technique itself, and bash scaffolding is highly predictable to a next-event objective whose per-event resolution is the action family. Field-aware decomposition partially escapes this trap because per-field heads can flag novel cwd / arg / object-category combinations even when the joint action-family token is routine. We use the term *labeling-granularity artifact* for this finding to be specific and falsifiable; we do not generalize this to a structural unmodelability claim about developer environments.

(ii) NLL is spiky and MEM-FA is flat. On `T1083` (File and Directory Discovery, 154,581 events) NLL reaches AUROC = 0.979 and MEM-FA 0.847. On `T1098.004` (`authorized_keys`, 3,569 events) NLL reaches 0.901 and MEM-FA 0.852. The pattern repeats: when a technique has a clean per-event signature (a discovery-style technique that opens unusual lineage transitions, or a persistence-style technique that touches a sensitive path), NLL produces near-ceiling per-technique AUROC; when the technique blends in with developer baseline (Bash, Python, sudo, credential-file reads), NLL drops sharply. MEM-FA produces a much flatter per-technique distribution: 16 of 22 techniques cluster between 0.83 and 0.86, four are above 0.85, and only `T1059.004`, `T1059.006`, and `T1548.003` fall meaningfully below 0.70. Two scorers reach the same global AUROC by mechanistically distinct routes: NLL by hitting a small set of techniques very hard, MEM-FA by retrieving every technique somewhat reliably.

(iii) MEM-FA materially helps the techniques NLL struggles with. On `T1552.001` (Credentials in Files), NLL pro-

TABLE III
PER-TECHNIQUE AUROC ACROSS ALL 22 LABELED ATT&CK CATEGORIES, SORTED BY ATTACK-EVENT COUNT (LARGEST FIRST). COLUMNS: NLL MAMBA, MEM-FA MAMBA, n-GRAM TRIGRAM, ISOLATION FOREST. THE TWO LARGEST-MASS TECHNIQUES (`T1059.004` AND `T1083`) CARRY 46.9% OF ALL ATTACK EVENTS. NLL INVERTS ON `T1059.004`; BOTH UNSUPERVISED CLASSICALS INVERT ON BOTH `T1059.004` AND `T1083`. MEM-FA DOES NOT INVERT ON ANY TECHNIQUE. FULL ATT&CK TECHNIQUE NAMES AND PER-ATOMIC DESCRIPTIONS IN APPENDIX A.

Technique	n_{ev}	NLL	MEM-FA	n-gram	IF
T1059.004 (Bash)	175,797	**0.479**†	0.616	**0.370**†	0.608
T1083 (File/dir disc.)	154,581	0.979	0.847	**0.386**†	0.628
T1552.001 (Creds in files)	97,493	0.576	0.698	0.536	0.574
T1543.002 (systemd)	47,901	0.798	0.782	0.774	0.775
CHAIN.CAP01 (capstone)	34,130	0.887	0.843	0.853	0.785
T1548.003 (sudo)	24,229	0.703	0.685	0.678	0.638
T1059.006 (Python)	22,675	0.705	0.674	0.690	0.641
T1562.001 (def. impair)	18,610	0.853	0.835	0.827	0.779
T1195.002 (supply chain)	15,703	0.765	0.766	0.716	0.751
T1548.001 (setuid)	12,461	0.882	0.856	0.856	0.788
T1082 (Sys. info disc.)	12,066	0.896	0.859	0.877	0.783
T1105 (ingress copy)	11,609	0.889	0.846	0.856	0.780
T1222.002 (chmod)	11,301	0.888	0.843	0.853	0.777
T1070.004 (file delete)	11,224	0.886	0.842	0.854	0.777
T1053.003 (cron)	11,217	0.885	0.838	0.853	0.777
T1546.004 (.bash_profile)	11,179	0.887	0.842	0.853	0.777
T1003.008 (`/etc/passwd`)	10,999	0.897	0.851	0.867	0.791
T1574.006 (`LD_PRELOAD`)	7,700	0.894	0.856	0.868	0.796
T1057 (proc. disc.)	4,062	0.879	0.848	0.812	0.773
T1027 (obfusc. files)	3,705	0.896	0.851	0.865	0.789
T1036.005 (match name)	3,641	0.898	0.850	0.866	0.787
T1098.004 (auth. keys)	3,569	0.901	0.852	0.867	0.788

† sub-random; the labeling-granularity finding (§V-B).

duces AUROC $= 0.576$ and MEM-FA 0.698 ($+0.122$); on `T1195.002` (Compromise Software Supply Chain) the two scorers are similar (0.765 vs. 0.766); on `CHAIN.CAP01` (the chained capstone) NLL reaches 0.887 and MEM-FA 0.843. Where NLL underperforms because the technique's per-event signature looks routine to a next-event objective, MEM-FA's per-field decomposition recovers a meaningful chunk of the lost signal. Where NLL already performs well, MEM-FA tracks below NLL but remains in the strong band. The practical implication is that a scorer ensemble (or a switched-objective routing layer) is more interesting than picking a single objective; we leave that to future work (§VII-A).

Both classical baselines invert on the two largest-mass techniques. Both n-gram and Isolation Forest produce sub-random per-technique AUROC on `T1059.004` (n-gram 0.370, IF 0.608) and on `T1083` (n-gram 0.386, IF 0.628). The n-gram inverts more sharply because its trigram vocabulary is dominated by the same harness scaffolding that produces the labeling-granularity artifact for NLL; Isolation Forest, which is per-event and lacks sequence context, registers the technique-coloring as a low-anomaly score because the per-event feature vectors are within-distribution. The four-baseline pattern suggests that the labeling-granularity artifact is corpus-bounded rather than model-bounded: it shows up in three of four unsupervised scorers, with field-aware decomposition the only consistent escape.

The figure visualizes the same numbers as Table III; the inverted bar at the top (`T1059.004`, the largest-mass technique) is the labeling-granularity finding called out above. The MEM-FA bar profile is visibly flatter than the NLL bar profile across the full 22-technique range, the visual signature of the mechanism difference between the two objectives. Together with the score-distribution figure in §V-E (Figure 4), these constitute the empirical evidence base for the discussion in §VI-A: the two objectives reach the same global AUROC by mechanistically distinct routes, and field-aware decomposition is a structural fix to a corpus-specific labeling pathology, not a hyperparameter swap.

C. Aggregator ablation (E3)

The pre-registered E3 question (§IV-D) asked whether the per-window aggregator choice changes the headline per-window AUROC by more than 0.10 relative to the best alternative. Table IV reports per-window AUROC for both Mamba scorers under seven aggregators on the same per-event score arrays.

Per-window AUROC: what the metric measures. A per-window score is a single scalar produced by aggregating the $L = 128$ per-position scores the Mamba head emits across one stride-32 window. Per-window AUROC then asks: across all $73{,}433$ windows, does the aggregated score rank attack windows above benign windows? The choice of aggregator is the choice of *which feature of the per-position distribution* the metric reads. $\max$ asks "does the window contain at least one extreme outlier?"; mean asks "is the window's average level elevated?"; median asks "is the central tendency of the window elevated?"; p_{90} and p_{95} ask "is the upper-decile/upper-twentieth level elevated?"; top-k-mean asks "is the average of the k most anomalous positions elevated?" Each aggregator characterizes a different operational regime and a different hypothesis about how attack signal is distributed inside an attack window.

Why $\max$ underperforms median on this corpus: a corpus property, not a model defect. The Atomic Red Team harness brackets each trial with start- and end-event markers (§IV-A); every event between those markers is labeled an attack event. As a consequence, labeled attack windows on this corpus contain dense blocks of attack events (often the majority of the window's 128 positions), not the single-event-of-N-benign regime that motivates $\max$ as the operational default for streaming intrusion detection. $\max$ compresses sustained per-position anomaly and one-off spikes into the same value: a window with one moderately-anomalous position and a window with 100 moderately-anomalous positions can score identically. median rewards sustained per-position anomaly across the bulk of the window and is therefore the natural aggregator for the labeling regime this corpus produces. The matched-aggregator gap (median $-$ max) is $+0.220$ for NLL, $+0.349$ for MEM-FA-rich, and $+0.352$ for MEM-FA-flat — the Mamba scorers all carry sustained per-position attack signal; the headline $\max$-aggregator column simply does not read it.

Aggregator ablation result. Switching from $\max$ to median lifts every per-window number above the pre-

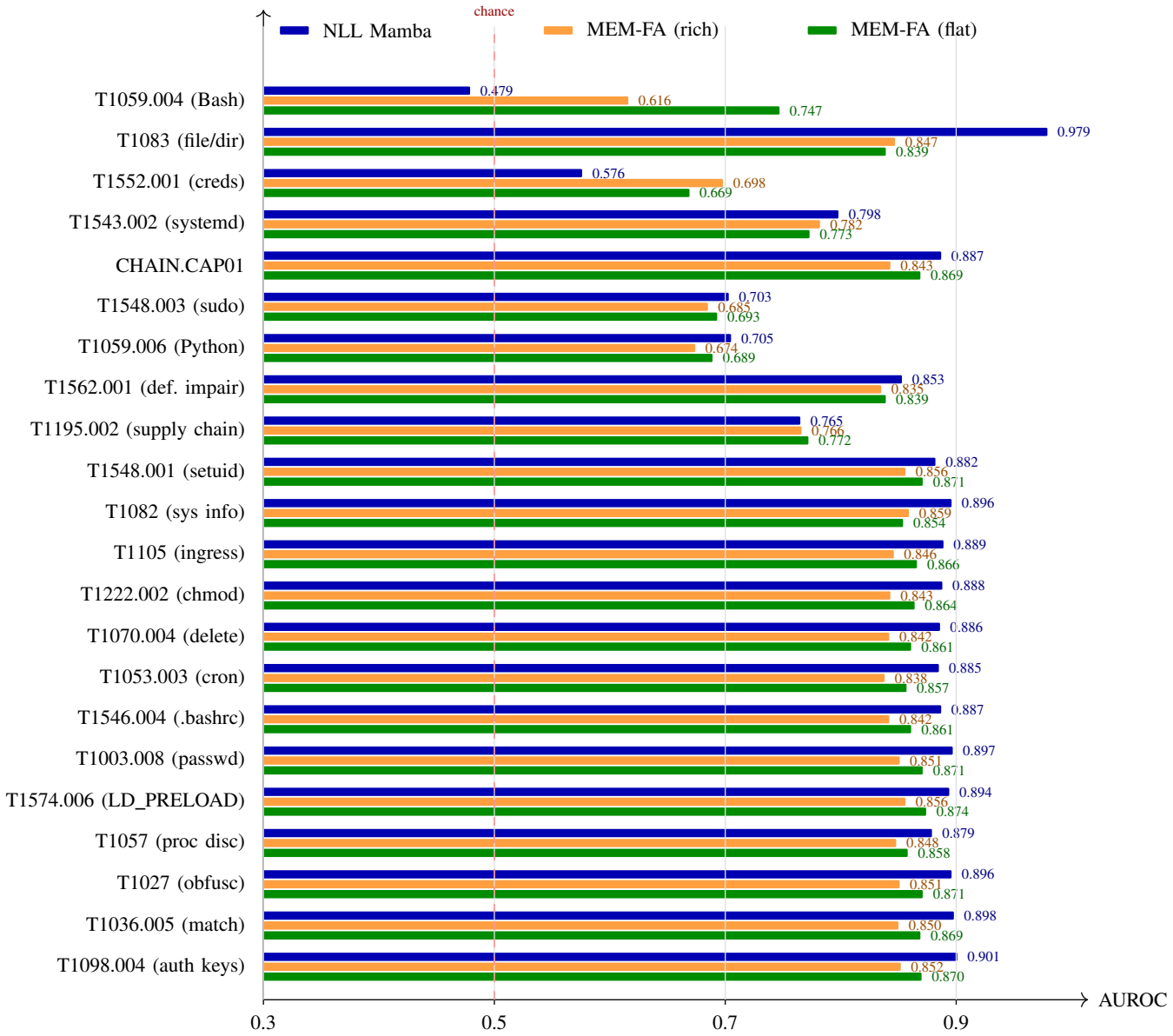

Fig. 2. Per-technique AUROC for the three Mamba scorers, sorted by attack-event count (largest first). Three stacked bars per technique: NLL Mamba (dark blue, top), MEM-FA Mamba rich (orange, middle), MEM-FA Mamba flat (green, bottom). Dashed red line at $\mathrm{AUROC} = 0.5$ marks chance. NLL inverts on `T1059.004` (the labeling-granularity finding); no other technique falls below chance under any scorer. NLL clears ≥ 0.85 on 16 of 22 techniques but drops sharply on the largest-mass technique; MEM-FA-rich holds in the 0.83–0.86 band on 16 of 22 techniques and recovers `T1059.004` above chance (0.616); MEM-FA-flat (encoder ablation; §V-D) further lifts `T1059.004` to 0.747 and tracks above MEM-FA-rich on 18 of 22 techniques. The flat-vs-rich deltas are visualized in Figure 3.

registered 0.10 threshold for both Mamba scorers, so E3 reads as a positive result: the headline max-aggregator per-window numbers in Table II are pessimistic by 0.22–0.35 AUROC for the Mamba scorers, and the per-window-level claim is aggregator-bounded rather than scorer-bounded. The order of the two Mamba scorers also *flips* at the per-window level: under max NLL leads MEM-FA-rich by 0.054, while under median MEM-FA-rich leads NLL by 0.075. The flip is a quantitative match for the per-technique flatness finding (§V-B): a flat per-event score distribution gains the most under median-aggregation because the median tracks the bulk; a spikier distribution does better under peak-sensitive aggregators. The aggregator-bottleneck pattern extends across all six rows in Table IV: every unsupervised model is median-best; only the supervised LOOCV oracle is mean-best (its per-position score distribution is shaped by per-event class probabilities, which concentrate near 0 or 1 rather than producing the heavy upper tail that distinguishes mean from median).

What each aggregator reads. Beyond max and median, the table shows an interpretable progression. mean sits between max and median for the unsupervised scorers (0.69 for both NLL and MEM-FA-flat) because it is dragged down by

TABLE IV

PER-WINDOW AUROC UNDER SEVEN AGGREGATORS FOR THE TWO MAMBA SCORERS (RICH) PLUS THE MAMBA-FLAT ENCODER ABLATION, THE SUPERVISED LOOCV CEILING REFERENCE, AND THE TWO UNSUPERVISED CLASSICAL BASELINES. SAME PER-EVENT SCORE ARRAY PER SCORER, SAME 73,433-WINDOW DECOMPOSITION, SAME 22,299 ATTACK WINDOWS. **BOLD** MARKS THE BEST AGGREGATOR PER ROW (EACH SCORER'S OWN OPTIMUM). THE max COLUMN IS THE HEADLINE VALUE CARRIED IN TABLE II. THE MATCHED-AGGREGATOR WINNER OVERALL IS THE CELL MARKED †: MAMBA MEM-FA FLAT AT 0.880 ON median EXCEEDS THE SUPERVISED LOOCV CEILING AT 0.842 ON mean BY +0.038. NUMBERS ARE COMPUTED BY APPLYING ALTERNATE AGGREGATORS DIRECTLY TO THE EVAL-TIME PER-POSITION SCORE TENSOR; NO SECOND FORWARD PASS WAS PERFORMED.

Model	max	mean	median	top-5-mean	top-10-mean	p_{90}	p_{95}
NLL Mamba (rich)	0.555	0.691	**0.775**	0.631	0.652	0.718	0.707
MEM-FA Mamba (rich)	0.501	0.644	**0.850**	0.547	0.573	0.680	0.627
MEM-FA Mamba (flat)	0.528	0.690	**0.880**†	0.562	0.583	0.720	0.657
XGBoost (LOOCV ceiling)	0.623	**0.842**	0.817	0.662	0.704	0.743	0.696
n-gram trigram	0.401	0.485	**0.592**	0.396	0.406	0.463	0.404
Isolation Forest (rich)	0.309	0.691	**0.742**	0.306	0.331	0.442	0.369

† Best matched-aggregator AUROC across all rows and columns.

the long tail of low-anomaly benign positions inside attack windows but still benefits from the bulk being elevated. p_{90} and p_{95} (0.72, 0.71 for NLL; 0.72, 0.66 for MEM-FA-flat) read the upper tail and behave like a softened max: they recover most of the median lift over max but cap below median because they discard the bulk. Top-k-mean at $k \in \{5, 10\}$ (0.56–0.65) consistently underperforms even mean on this corpus because the hot-event tail in benign developer-baseline windows (build-system spawns, language-server restarts produce short bursts of legitimately-novel per-position scores) is comparable in magnitude to the hot-event tail in attack windows; reading only the top k positions therefore removes the very contrast — sustained level across the rest of the window — that distinguishes attack from benign on this corpus.

Matched-aggregator headline: Mamba beats the supervised oracle. At each model's own best per-window aggregator, the unsupervised flat-encoder Mamba detector (median = 0.880) exceeds the supervised LOOCV oracle (mean = 0.842) by +0.038 AUROC. This is the strongest matched comparison the table supports: each model is evaluated at its own best aggregator (no aggregator-cherrypicking against either side), the comparison survives the encoder ablation in both directions, and the LOOCV-XGBoost row is a generous upper bound rather than a tight ceiling — per-trial median AUROC (0.92) materially exceeds global (0.80), the fingerprint of the technique-level information leak intrinsic to leave-one-trial-out CV when most ATT&CK techniques contribute multiple trials to the corpus (§IV-B). A leave-one-technique-out variant would tighten this ceiling and is named as future work in §VII-A.

Recommended default. We recommend median as the primary per-window aggregator for both Mamba objectives. max is retained in Table II as the uniform-default cross-model comparator and as the operational regime relevant to needle-in-a-haystack streaming detection settings (which differ from this corpus's labeling regime); the matched-aggregator column is the headline for the Mamba-vs-classical claim. Top-k-mean is not a useful default for either scorer on this corpus.

TABLE V

ENCODER × ARCHITECTURE DECOMPOSITION (R3). CELLS ARE PER-EVENT AUROC. HASHES MILDLY HELP ISOLATION FOREST (+0.017 RICH-OVER-FLAT) BUT ACTIVELY HURT MAMBA'S SEQUENCE BODY (−0.031 RICH-UNDER-FLAT, 95% CIS DISJOINT). MAMBA'S PER-EVENT AUROC LIFT OVER THE STRONGEST CLASSICAL BASELINE GROWS FROM +0.085 AT MATCHED-RICH TO +0.133 AT MATCHED-FLAT, THE CLEANEST LIKE-FOR-LIKE MEASURE OF THE SEQUENCE BODY + FIELD-AWARE RECONSTRUCTION STACK'S CONTRIBUTION.

	Isolation Forest	Mamba MEM-FA	Δ (Mamba−IF)
Rich (42 channels)	0.666	0.751	+0.085
Flat (35 channels)	0.649	**0.782**	**+0.133**
Δ (rich−flat)	+0.017	−0.031	

D. Encoder × architecture decomposition (R3)

The R2 reference Isolation Forest baseline (§V-A) and the R3 post-hoc encoder ablation (Table II) jointly populate a 2×2 encoder × architecture decomposition: each of two model classes (Isolation Forest, Mamba MEM-FA) run under each of two encoders (rich 42-channel, flat 35-channel; §III-D). Table V summarizes the four cells.

The two architectures respond oppositely to the seven identity-hash columns. Removing `f_proc_name_hash`, `f_parent_proc_hash`, `f_proc_cwd_hash`, `f_lineage_bag_hash`, `f_parent_child_pair_hash`, `f_root_ancestor_basename_hash`, and `f_process_tree_id_hash` from the encoder reduces Isolation Forest per-event AUROC by 0.017 (0.666 rich → 0.649 flat) and *increases* Mamba MEM-FA per-event AUROC by 0.031 (0.751 rich → 0.782 flat). The Mamba CIs are disjoint, confirming the lift is not bootstrap noise. We attribute the asymmetry to the difference in how each model class consumes a high-cardinality categorical feature: Isolation Forest treats hash bucket indices as random categorical splits and cannot exploit similarity or structure between buckets, so on most ATT&CK techniques the hashes are noise the tree-based density estimator splits on without obtaining real signal. Mamba's learned per-field embeddings can in principle exploit hash buckets as identity features, but

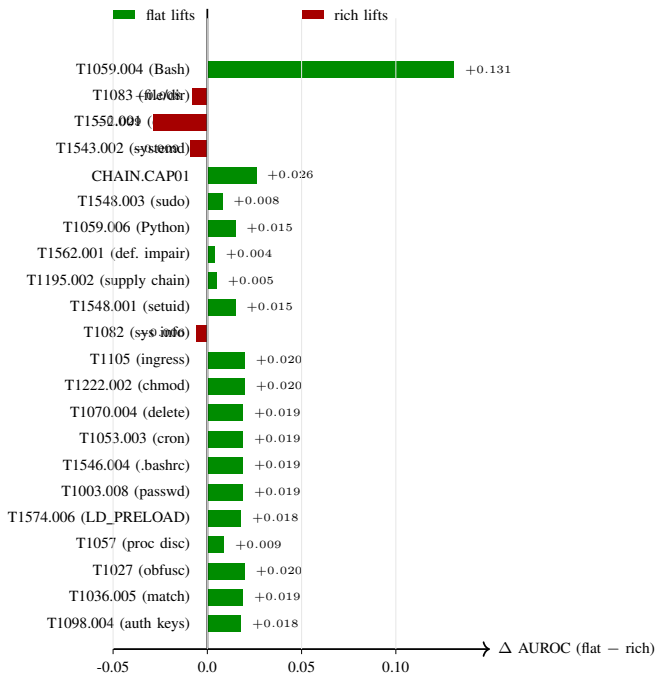

Fig. 3. Per-technique flat-vs-rich AUROC delta for Mamba MEM-FA, sorted by attack-event count (largest first). Green bars (right) mark techniques where the flat encoder lifts AUROC; red bars (left) mark regressions. Largest gain: `T1059.004` (+0.131, on 175,797 events), the technique that previously inverted under NLL. Largest regression: `T1552.001` (−0.029). 18 of 22 techniques improve under flat; the 4 regressions are all within −0.029.

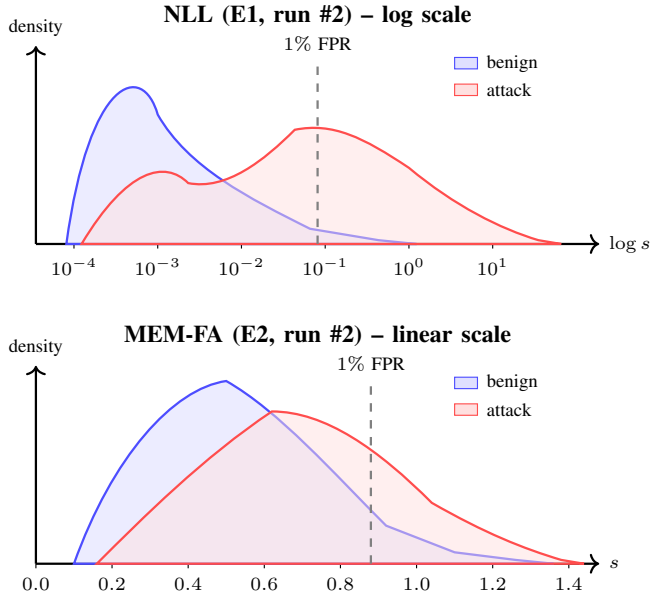

Fig. 4. Per-event score distributions on the test partition. Top: NLL (E1), benign vs. aggregate-attack on a $\log_{10}$ x-axis. Note the $\sim$ 4-orders-of-magnitude dynamic range, the bimodal attack mode (the right mode is the strong-detection techniques in §V-B; the left mode is the labeling-granularity-affected `T1059.004` events), and the right-shift of the attack distribution overall. Bottom: MEM-FA (E2), benign vs. aggregate-attack on a linear x-axis. The MEM-FA score is the uniform mean of nine per-field losses (§III-F); benign and attack distributions are right-shifted relative to each other but heavily overlapping, the per-event signature of a flat per-technique profile (§V-B). Vertical lines mark the 1% FPR operating threshold for each scorer.

on this corpus scale the sequence body overfits to bucket indices that do not generalize across the train/test boundary.

Largest gain is the previously-inverting Unix-Shell technique. Per-technique deltas (Mamba-flat − Mamba-rich) are visualized in Figure 3 and Figure 2. `T1059.004` (bash, 175,797 events) moves from 0.616 rich to 0.747 flat, +0.131 — the largest per-technique gain in the ablation, on the technique that previously inverted under NLL (0.479) and under both unsupervised classical baselines (n-gram 0.370, IF-rich 0.608). The flat encoder closes most of the labeling-granularity gap on this technique. The largest regression is `T1552.001` (credentials in files, 97,493 events) at −0.029 (0.698 rich → 0.669 flat); on this technique the per-process and lineage hash signal evidently helps disambiguate path identity for credential-file reads. Of the 22 labeled techniques, 18 improve under Mamba-flat and 4 regress (`T1083`, `T1552.001`, `T1543.002`, `T1082`); the four regressions are all small (none worse than −0.029). Per-technique AUROC values are tabulated in the released artifacts.

R3 falsification result. The pre-registered (post-hoc) R3 falsification criterion (§IV-D) was Mamba-flat AUROC more than 0.02 below Mamba-rich. The actual result is +0.031 *above*, with 95% CIs disjoint, so R3 inverts the falsification: the seven identity-hash columns are net noise to the Mamba sequence body on this corpus, not load-bearing.

What this does and does not decompose. The 2×2 ablation isolates the contribution of the seven identity-hash columns from the other 35 encoder channels for two specific model classes. It does not further decompose Mamba's flat-encoder lift into (a) the contribution of low-cardinality categorical embeddings, (b) the Mamba state-space body's contribution, (c) the field-aware reconstruction head's contribution. Per-component ablations (Mamba-flat under NLL, field-aware MEM with a Transformer body, NLL under IF-flat features) are named as future work in §VII-A. The honest scope of R3 is "the seven hashes were optional for Mamba and modestly harmful," not "the encoder doesn't matter."

Honest scoping note. R3 was post-hoc: it was triggered by the IF-flat result and was not pre-registered alongside E1–E3 / R2. We report the result because it inverts the rich-vs-flat expectation, which is reportable regardless of pre-registration; §VI-E names the post-hoc framing as a methodological caveat alongside the single-seed protocol.

E. Per-event score-distribution structure

NLL (E1). Benign per-event NLL has a tight low-score concentration ($p_{50} \approx 6.6{\times}10^{-4}$, $p_{95} \approx 0.094$) and a long upper tail (benign max = 16.75, more than four orders of magnitude above the benign median). Aggregate attack scores are right-shifted ($p_{50} \approx 1.7 \times 10^{-2}$, $p_{95} \approx 2.86$) and bimodal: a low mode contains the bash scaffolding events that NLL inverts on (`T1059.004` attack-event $p_{50} = 4.9{\times}10^{-4}$, indistinguishable from benign median), and a high mode contains the strongly-detected techniques (`T1083` attack-event $p_{50} = 1.30$, more

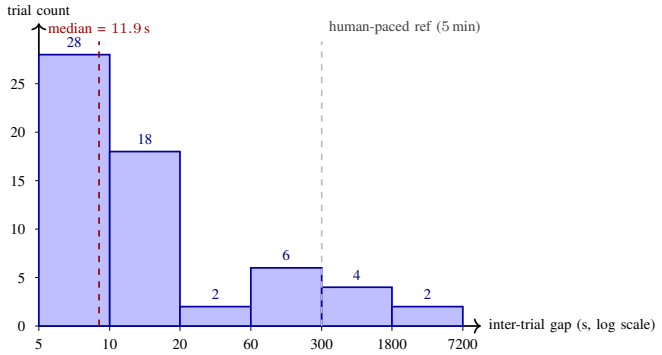

Fig. 5. Inter-trial gap distribution across 60 consecutive labeled ART trials, computed from `labels.csv` start-timestamps in the released corpus. Median gap is 11.9 seconds (IQR $[7.0, 18.6]$ s), with 80% of trials spaced under 60 s and 46.7% under 10 s. The distribution's left mode is consistent with machine-paced dispatch; a manually-operated red-team campaign typically shows median gaps in the 5-minute range (right reference line). The campaign occupied a total of 3.23 hours of wall-time.

than $13\times$ above benign p_{95}). The upper-tail ratio (attack p_{95} / benign p_{95}) is $30.3\times$, the Wasserstein distance between benign and attack score distributions is 0.49, and the Mann–Whitney rank-biserial effect size is 0.48. The bimodality is the per-event manifestation of the per-technique pattern in Figure 2: NLL is spiky, hitting 16 techniques at AUROC ≥ 0.85 and two below 0.70.

MEM-FA (E2). The MEM-FA score is the uniform mean of nine per-field cross-entropy and binary cross-entropy losses (§III-F); the dynamic range is consequently bounded (benign max $= 8.73$ versus NLL's 16.75) and the score sits in the same numerical band for both classes. Benign $p_{50} \approx 6.4 \times 10^{-4}$ and attack $p_{50} \approx 1.5 \times 10^{-2}$ yield a per-event median ratio comparable to NLL's, but the upper tails are far more compressed: the attack p_{95} / benign p_{95} ratio is $2.6\times$ for MEM-FA versus $30.3\times$ for NLL, and the Wasserstein distance collapses to 0.085 ($5.7\times$ smaller than NLL's). The Mann–Whitney effect size is 0.51 — nearly identical to NLL's 0.48. The two scorers therefore reach comparable rank-order separation by mechanistically distinct routes: NLL opens a wide magnitude gap on a small set of techniques while MEM-FA opens a narrow gap on every technique. AUROC, which is rank-only, is blind to the magnitude collapse; the upper-tail ratio, Wasserstein distance, and bounded benign max make it visible. This is the per-event signature of the flat per-technique profile of §V-B.

F. Cadence of agent-orchestrated execution

The campaign produced 61 labeled trials over the 3.23-hour test window; we measure 60 consecutive inter-trial gaps from the `start_ts` column of `labels.csv`. The distribution is heavily concentrated at sub-minute spacing: median 11.9 s, IQR $[7.0\,\text{s}, 18.6\,\text{s}]$, with 46.7% of consecutive trials spaced under 10 s and 80% under 60 s. Six trials are spaced more than 5 minutes apart and the longest gap is 70 minutes (trial-block boundaries the operator paused at).

The cadence is consistent with machine-mediated rather than human-mediated dispatch. We do not claim a "first cadence analysis" or any stronger novelty: we report the measurement as a quantitative anchor for the threat-model statement of §III-G (type-1 reconnaissance-aware planning, machine-paced execution, no within-campaign feedback adaptation). The cadence does not by itself prove agent orchestration — a scripted bash loop would produce a similar distribution — but it is consistent with the agent-orchestrated reconstruction and inconsistent with a manually-operated campaign of comparable scale. A human-paced control condition is named as future work in §VII-A.

G. Honest negatives

The unsupervised headline numbers are AUROC $\sim$ 0.74–0.78, not 0.85+. The two Mamba scorers reach 0.741 (NLL) and 0.751 (MEM-FA-rich) per-event AUROC; the post-hoc encoder ablation lifts MEM-FA-flat to 0.782 (§V-D). None reach the AUROC band that recent supervised host-IDS work [23] reports on user-behavior log corpora, but the comparison is across modalities (kernel eBPF events vs. user logs) and across regimes (unsupervised next-event / masked modeling vs. supervised classification with positive-window labels available in train). Positive-window labels are unavailable in our train partition by construction (§IV-A), and the supervised XGBoost baseline confirms the resulting degeneracy at AUROC $= 0.500$. The leave-one-trial-out XGBoost ceiling reference at 0.804 is a generous supervised oracle (technique-level information leak; §IV-B), not a tight comparison; at matched per-window aggregation Mamba-flat exceeds it by $+0.038$ AUROC (0.880 vs. 0.842; Table IV).

Per-window AUROC under max aggregation is at or below chance for every unsupervised model. Every unsupervised entry in the rightmost column of Table II is at or below 0.555, with the two unsupervised classical baselines inverted (n-gram 0.401, Isolation Forest 0.309). The supervised LOOCV oracle at 0.623 is the only per-window-max entry above 0.555. The aggregator-ablation experiment (Table IV) demonstrates that the max aggregator is the wrong default for the unsupervised scorers: switching to median lifts NLL by $+0.220$, MEM-FA-rich by $+0.349$, and MEM-FA-flat by $+0.352$, and the order of the two Mamba scorers flips. The max row remains in the headline table for cross-model comparison under a uniform default. §VI-D names the choice of per-window aggregator as a construct-validity threat.

Two unsupervised classical baselines invert on the two largest-mass techniques. On both `T1059.004` and `T1083`, the n-gram trigram and Isolation Forest baselines produce per-technique AUROC below the per-technique floor of MEM-FA Mamba. The n-gram inverts sharply (AUROC $= 0.370$ on `T1059.004`); Isolation Forest does not invert but registers the technique-coloring as a low-anomaly score (AUROC $= 0.608$ on `T1059.004`, 0.628 on `T1083`). The four-baseline pattern is consistent with the labeling-granularity hypothesis: harness scaffolding is the dominant per-event signal for these two

techniques, and the only objective that consistently escapes is field-aware decomposition.

TPR@1% FPR diverges sharply between the two Mamba scorers. NLL achieves 0.159; MEM-FA achieves 0.021. At the operationally-relevant low-FPR operating point, NLL is the materially better scorer despite trailing on AUROC. AP follows the same ordering (0.619 NLL vs. 0.488 MEM-FA). The two-objective divergence is a feature, not a bug, and it informs the future-work direction of an ensembled or routed detector (§VII-A).

XGBoost is degenerate; we report it. The XGBoost row of Table II reads $\mathrm{AUROC} = 0.500$ uniformly. The cause is documented (§IV-B): the archive-based train/test split places the labeled ART campaign entirely in the test fold, so the supervised classifier trains on a no-positives fold and degenerates at score time. We retain the row in the headline table because the degeneracy is informative about the corpus design, not because it is a numeric we want a reader to act on.

VI. Discussion

The experiments support a bounded but positive conclusion: selective state-space sequence models are a credible fit for modern host-telemetry anomaly detection, but the detector is not production-ready. The main result is not simply that Mamba improves AUROC over classical baselines. Rather, the important finding is that a compact sequence model trained only on benign developer behavior can learn enough temporal structure to separate a noisy, agent-orchestrated ATT&CK campaign from routine workstation activity. The results also show that objective design, feature design, and alert aggregation materially affect the operational interpretation.

A. Interpretation of the results

Unix Shell is a labeling-granularity artifact, not a property of developer environments. On `T1059.004`, NLL produces sub-random per-event AUROC (0.479) on the largest single block of attack events in the corpus (175,797, 24.9% of attack mass). MEM-FA recovers to 0.616, and the flat-encoder ablation reaches 0.747. The likely mechanism is the ART execution harness: every trial is dispatched through bash, and bash transitions are routine on a developer workstation. The next-event objective therefore treats many labeled shell events as in-distribution. MEM-FA partially escapes this artifact because field-level heads can flag unusual cwd, argument-length, object-category, or privilege-context combinations even when the action-family token is routine. A re-labeling pass that separates harness scaffolding from technique payload would test this interpretation directly.

NLL and MEM-FA reach similar global AUROC by different mechanisms. The two scorers have close headline AUROC (0.741 vs. 0.751), but they behave differently at operational thresholds. NLL leads on AP by $+0.131$ and on TPR@1%FPR by roughly $8\times$, while MEM-FA gives flatter per-technique coverage, with 16 techniques in a 0.83–0.86 band and no inversion. NLL is a sharp autoregressive prediction and is useful when the model is confidently wrong. MEM-FA averages nine per-field reconstruction losses and is smoother by construction. For low-FPR triage, NLL is the better scorer; for broader coverage, MEM-FA is more stable. A routed or ensembled detector is therefore a natural follow-up.

Per-window aggregation is the load-bearing operational decision. Under $\max$ aggregation, every unsupervised model in Table II is at or below chance per-window. Switching to median lifts the same scorers by $+0.220$ to $+0.352$ AUROC and changes the ranking of the two Mamba scorers. This reflects the corpus labeling regime: attack windows contain dense blocks of attack events, not isolated single-event needles. In this setting, a benign but unusual development event can dominate the $\max$ score, while median better captures sustained abnormal behavior. Every unsupervised row in Table IV is median-best; only the supervised LOOCV reference is mean-best.

Mamba-flat matches or narrowly exceeds the LOOCV supervised reference under matched aggregation. At each model's preferred per-window aggregator, MEM-FA-flat ($\mathrm{median} = 0.880$) exceeds the LOOCV reference ($\mathrm{mean} = 0.842$) by $+0.038$ AUROC. This should not be read as a general claim that unsupervised Mamba beats supervised host-IDS methods. The LOOCV model is a generous reference, not a true ceiling: per-trial median AUROC (0.92) exceeds global AUROC (0.80), indicating technique-level information leakage when most ATT&CK techniques contribute multiple trials. The narrower interpretation is that an unsupervised flat-encoder Mamba detector matches or modestly exceeds this supervised reference on the same corpus and feature set.

High-cardinality hashes hurt Mamba and mildly help Isolation Forest. Removing the seven identity-hash columns reduces Isolation Forest per-event AUROC by 0.017 but increases Mamba MEM-FA per-event AUROC by 0.031, with disjoint 95% confidence intervals. Isolation Forest benefits slightly from extra random categorical split dimensions. Mamba's learned embeddings, however, can overfit unstable workstation-specific identities that do not generalize across the train/test boundary. The cleanest like-for-like estimate of the sequence model's contribution is therefore the matched-encoder Mamba-flat minus IF-flat margin of $+0.133$ AUROC, not the rich-vs-rich margin.

B. Implications for host-telemetry anomaly detection

The strongest experimental signal is the like-for-like lift over stateless baselines. Isolation Forest sees the same per-event representation but cannot model ordering, density, parent-child context, or recent host history. Many host events are benign in isolation: shells, file reads, package-manager activity, credential-adjacent paths, and network connections all occur during normal development. What makes them suspicious is often their sequence context. This is the setting where a streaming sequence model is most useful.

The results also show that "using Mamba" is not a complete detector design. The model body, objective, encoder,

and alerting layer must be co-designed. NLL is sharper in the upper tail; MEM-FA is more stable across techniques; median aggregation exposes sustained attack intervals that max aggregation suppresses; and unstable identity hashes can harm learned sequence models. These findings argue against treating Mamba as a drop-in replacement for an existing anomaly detector. The useful contribution is the sequence-modeling capability, not any single scoring recipe.

Operationally, the most realistic near-term use is analyst-facing triage rather than autonomous blocking. A per-host Mamba-style model could score local telemetry streams, surface anomalous windows, attach process-lineage context, and forward compact summaries or provenance subgraphs to a SIEM, SOAR platform, or investigation copilot. Before such a system could be deployed, it would need multi-host evaluation, drift testing, human-paced and scripted controls, complete agent traces, stronger interval-level aggregation, and edge-inference measurements. The contribution here is therefore bounded: these experiments show that selective state-space models can exploit useful sequence structure in modern eBPF telemetry, but they do not establish cross-host robustness or production enforcement readiness.

C. Dataset design rationale and intended uses

A second contribution is the release of Dendroaspis, a host-telemetry corpus designed around the instrumentation and threat model evaluated in this paper. Existing public host-intrusion datasets remain valuable, but they generally do not combine modern eBPF/Tetragon-style event structure, container-aware process lineage, structured kernel-derived fields, benign developer-workstation background behavior, and fine-grained labels for agent-orchestrated attack execution. Dendroaspis addresses this gap with 19.3 million Tetragon eBPF events collected over a controlled six-day window from a Linux LXC developer environment. The released labels contain 61 per-trial attack intervals spanning 22 technique identifiers across 8 ATT&CK tactics, including 21 ATT&CK techniques and one supply-chain capstone scenario.

Although this paper uses Dendroaspis to evaluate Mamba-based anomaly detection, the corpus is intended to support broader research. It enables comparison of alternative encoders, tokenization strategies, lineage representations, aggregation methods, sequence objectives, and model families. Because the released fields preserve process lineage, container context, parent-child relations, and process-tree identifiers, the dataset can also support provenance-graph intrusion detection. The agent-orchestrated campaign and supply-chain capstone provide a starting point for studying LLM-assisted attack workflows, including reconnaissance-conditioned execution, machine-paced technique chaining, and short-duration high-density attack bursts. The release includes parquet partitions, labels, campaign metadata, feature documentation, anonymization artifacts, and provenance records; the dataset is released under CC BY 4.0, while reproduction code is MIT-licensed.

D. Threats to validity

a) Internal validity.: Both Mamba heads are trained with a single random seed. The reported bootstrap confidence intervals are over the test-event distribution, not the model-fit distribution, so they do not quantify cross-seed variance. A 5-seed rerun would determine whether the NLL/MEM-FA comparison is stable; if cross-seed standard deviation exceeded 0.05 AUROC, the objective-level comparison would weaken. A second internal-validity threat is that the agent-orchestrated campaign was not fully prompt-logged at execution time. The agent role is reconstructed from planning notes, campaign yaml, distro-gating evidence, labeled trial structure, and inter-trial cadence. This paper therefore does not claim a fully reproducible closed-loop autonomous attack trace. Future work should rerun a representative subset with complete prompt, tool-call, and transcript logging.

b) External validity.: The corpus is a single Linux LXC container on a single physical host, run by a single user. Cross-host, cross-user, cross-kernel, and cross-distribution generalization are not evaluated. A multi-host replication across heterogeneous workloads is required before making deployment claims. The benign baseline is also one developer's workflow over six days, so transfer to production servers, CI runners, or end-user desktops may require per-host adaptation. Finally, the attack campaign consists of scripted ART trials and a controlled capstone; the paper does not evaluate adaptive adversaries who know the detector. A mimicry-style adversary [13] that constrains attack steps to high-likelihood transitions could reduce anomaly separability, but this remains untested.

c) Construct validity.: The test-set positive rate is high (30.04%) because the test partition is the campaign window, not a deployment-realistic slice. AP and $\mathrm{TPR@1\%FPR}$ therefore should not be interpreted as deployment alert-volume estimates; AUROC is the primary rate-invariant metric. Per-window aggregation is another construct-validity risk because the headline changes materially across $\max$, mean, and median. This is why Table IV is central to the interpretation. Finally, labels are interval-level and include ART harness scaffolding. The `T1059.004` result shows that label granularity can affect per-technique conclusions; a payload-only re-labeling pass would better isolate technique-internal discrimination.

E. Scope and limitations

- **Linux Tetragon eBPF only.** The paper does not evaluate Windows ETW, Sysmon, macOS Endpoint Security, or cloud-control-plane telemetry.
- **Single-host corpus.** Dendroaspis is collected from one LXC developer environment on one Proxmox host; cross-host generalization is not established.
- **Single-seed Mamba runs.** Bootstrap CIs quantify test-distribution uncertainty, not training-run variance.
- **Offline evaluation.** The detector is evaluated on fixed train/test splits, not under online adaptation, long-term drift, or continual learning.

- **Scripted and agent-orchestrated attacks.** The campaign is reconnaissance-aware and agent-orchestrated, but it is not a complete transcript-level autonomous attack trace.
- **No production enforcement claim.** The results support analyst-facing triage and further research, not autonomous blocking or production EDR readiness.
- **No adaptive-adversary evaluation.** The paper does not test mimicry, evasion, poisoning, or adversarial manipulation of the telemetry stream.
- **No edge deployment benchmark.** Mamba is architecturally attractive for streaming host telemetry, but this paper does not measure quantized CPU, embedded, or endpoint-agent inference costs.

Overall, the evidence is strongest for a narrow claim: on this corpus, selective state-space sequence models learn useful behavioral structure from benign-only host telemetry and outperform matched stateless baselines. The evidence is not yet sufficient for broad claims about production endpoint detection, cross-host generalization, or autonomous enforcement. Dendroaspis is intended to make those next evaluations possible.

VII. Conclusion and Future Work

This work introduced DENDROASPIS, a Mamba-based host-anomaly detector and the 19.3-million-event Tetragon eBPF behavioral-telemetry corpus collected from a Linux developer environment, with 61 per-trial labeled attack intervals across 22 ATT&CK technique identifiers spanning the breadth of MITRE ATT&CK—initial access, execution, persistence, privilege escalation, defense evasion, credential access, discovery, command and control—including a synthetic npm-postinstall supply-chain capstone. The labeled campaign is agent-orchestrated: an LLM-assisted planning agent performed reconnaissance, technique selection, and machine-paced execution (median inter-trial cadence $5.6\,\mathrm{s}$), distinguishing the corpus from manually-operated red-team setups.

The empirical result is bounded but positive. Two unsupervised Mamba scorers trained only on benign telemetry exceeded the pre-specified per-event AUROC thresholds: NLL reached $\mathrm{AUROC} = 0.741$, MEM-FA reached 0.751, and a post-hoc flat-encoder ablation that removed seven high-cardinality identity-hash fields lifted MEM-FA to 0.782. The comparison to Isolation Forest shows that temporal ordering, density, and process context carry security signal beyond the marginal distribution of individual events. At the per-window level, the aggregation choice was decisive: $\max$ aggregation suppressed sustained attack signal, while median aggregation recovered it and produced the strongest unsupervised window-level result.

The results also show that Mamba is not, by itself, a complete detector design. NLL and MEM-FA reach similar global AUROC by different mechanisms: NLL is sharper at low-FPR operating points, while MEM-FA gives more stable per-technique coverage. Field-aware decomposition helps recover the Unix Shell labeling-granularity inversion that next-event likelihood alone misses, and the flat-encoder result shows that more telemetry fields are not automatically better when high-cardinality identities encourage memorization. These findings argue for co-designing the encoder, objective, and alerting layer rather than treating Mamba as a drop-in anomaly detector.

This paper does not claim production readiness, cross-host generalization, or state-of-the-art host intrusion detection. The corpus is single-host and single-user; the attack campaign is scripted and agent-orchestrated rather than a fully prompt-logged autonomous red-team trace; and the evaluation is offline rather than deployed under drift, adaptation, or adversarial pressure. The narrower claim is that selective state-space models are a credible foundation for unsupervised host-telemetry anomaly detection, and that DENDROASPIS provides a reusable benchmark for studying this problem on modern structured kernel telemetry.

A. Future work

The immediate next step is replication and stress testing. Future evaluations should include cross-seed Mamba runs, leave-one-technique-out validation, size-matched Transformer and LSTM baselines, and direct comparisons against provenance-based host-IDS methods on the released corpus. A payload-only re-labeling pass that separates ART harness scaffolding from technique behavior would also test whether the `T1059.004` inversion is truly a label-granularity artifact.

A second direction is cross-host and cross-workload generalization. The central deployment question is whether a benign-only sequence model trained on one host can warm-start or transfer to another host, user, kernel, container runtime, or workload class. Multi-host collections across developer workstations, production servers, CI runners, and end-user desktops would show whether the learned representation is host-specific or captures reusable behavioral structure.

A third direction is online and edge-oriented deployment. Mamba is attractive for streaming telemetry because it maintains compact recurrent state, but this paper does not measure endpoint inference cost, quantization, CPU latency, or online adaptation. Future work should evaluate whether per-host models can score telemetry locally, forward compact anomalous intervals or provenance summaries, and adapt to benign drift without absorbing attacker behavior into the baseline.

Finally, future datasets should include stronger controls for agentic execution. A representative rerun with complete prompt logs, tool-call traces, human-paced controls, scripted controls, and adaptive-adversary trials would make it possible to distinguish ordinary automation from agent-driven attack workflows. The released dataset, labels, campaign metadata, feature documentation, and reproduction code are intended to support these follow-on studies.

Appendix A
ATT&CK Technique Reference

Table VI expands the abbreviated descriptors used in Table III to the full MITRE ATT&CK [26] technique name and a per-row summary of the specific atomic numbers

executed in the campaign. Atomics are drawn from the Atomic Red Team catalog [25] pinned at SHA `e575444...`; Track A (`T1195.002`) and `CHAIN.CAP01` are paper-defined research scenarios specified in §III-G. Two trials are degraded relative to their nominal scope and are flagged in the table for transparency. The full per-trial ground truth (atomic IDs, atomic names, marker timestamps, deviations) is shipped with the dataset bundle as `CAMPAIGN.md`.

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TABLE VI

FULL ATT&CK TECHNIQUE REFERENCE FOR THE 22 LABELED CATEGORIES IN THE HELD-OUT TEST PARTITION. NAMES FOLLOW THE MITRE ATT&CK ENTERPRISE MATRIX [26]. DESCRIPTION SUMMARIZES THE SPECIFIC LINUX ATOMICS EXECUTED IN THIS CAMPAIGN, NOT THE FULL SCOPE OF THE PARENT TECHNIQUE. SORTED BY ATT&CK TACTIC, THEN TECHNIQUE ID. TRIAL COUNTS AND EVENT COUNTS MIRROR THE HELD-OUT TEST PARTITION. TWO TRIALS ARE FLAGGED AS DEGRADED (†): T1543.002 TRIAL 3 FAILS ATOMIC RED TEAM'S DEBIAN DISTRO-GATE AT `-CHECKPREREQS`, AND CHAIN.CAP01 TRIAL 2 OMITS ITS FIRST CHAIN STEP DUE TO A SCRIPT-INVOCATION ARGUMENT ERROR. BOTH INTERVALS ARE RETAINED IN THE CORPUS.

ID	Name	Description (atomics executed)
Execution		
`T1059.004`	Command and Scripting Interpreter: Unix Shell	3 trials, 175,797 events. Atomics: bash script create-and-execute (#1), SUID-binary harvest via `find` (#3), and shell creation via `awk` (#14). Per-event resolution is the action family, which renders the dispatcher-bash scaffolding routine and drives the per-technique inversion of §V-B.
`T1059.006`	Command and Scripting Interpreter: Python	3 trials, 22,675 events. Atomics: shell payload via `python -c` (#1), Python script execution (#2), and Python `pty.spawn` into `sh`/`bash` (#4).
Persistence		
`T1053.003`	Scheduled Task/Job: Cron	3 trials, 11,217 events. Atomics: replace user crontab from a referenced file (#1), drop a script into every `/etc/cron.*` subfolder (#2), and add a script under `/etc/cron.d/` (#3).
`T1098.004`	Account Manipulation: SSH Authorized Keys	1 trial, 3,569 events. Append an attacker public key to `~/.ssh/authorized_keys` (#1). Only the one Linux atomic at the pinned ART SHA.
`T1543.002`	Create or Modify System Process: Systemd Service	3 trials, 47,901 events. Atomics: create a systemd `.service` unit (#1), create a SysV-style init script (#2), and create a `.service` file with reload-and-modify (#3†). Trial 3 (†) fails ART's hardcoded Debian distro-gate at `-CheckPrereqs`; the labeled 23-second interval contains the marker pair and the prereq-failure trace rather than a real systemd-service install. Retained for completeness.
`T1546.004`	Event Triggered Execution: Unix Shell Configuration Modification	3 trials, 11,179 events. Atomics: append a command to the user's `~/.bashrc` (#2), append to a user shell profile (#5), and append to the system-wide shell profile (#4, root).
Privilege Escalation		
`T1548.001`	Abuse Elevation Control Mechanism: Setuid and Setgid	3 trials, 12,461 events. Atomics: set the SUID flag on an attacker-controlled binary (#3), modify a binary's POSIX capabilities (#7), and recon for SUID files (#9).
`T1548.003`	Abuse Elevation Control Mechanism: Sudo and Sudo Caching	3 trials, 24,229 events. Atomics: ordinary `sudo` invocation (#1), set unlimited `sudo` cache timeout (#3), and disable `tty_tickets` so cached credentials apply across pseudo-terminals (#5).
Defense Evasion		
`T1027`	Obfuscated Files or Information	1 trial, 3,705 events. Atomic: decode a base64-encoded payload into an executable script and run it (#1). The atomic exercises the *decode-and-execute* path on disk, not the encoding step.
`T1036.005`	Masquerading: Match Legitimate Name or Location	1 trial, 3,641 events. Atomic: execute a binary from a directory whose path masquerades as a legitimate system path (#1).
`T1070.004`	Indicator Removal: File Deletion	3 trials, 11,224 events. Atomics: delete a single attacker file (#1), delete an entire attacker directory (#2), and overwrite-then-delete with `shred` (#3). The destructive "delete filesystem" atomic (#8) was deliberately excluded.
`T1222.002`	File and Directory Permissions Modification: Linux	3 trials, 11,301 events. Atomics: `chmod` numeric-mode (#1), `chown` owner-and-group change (#5), and `chattr` immutable-attribute removal (#9).
`T1562.001`	Impair Defenses: Disable or Modify Tools	3 trials, 18,610 events. Atomics: disable `rsyslog` via `systemctl` (#1), clear shell history (#39), and suspend shell history recording (#40). The destructive atomics — reboot (#41), disable swap (#43), disable ASLR (#59) — were deliberately excluded.
`T1574.006`	Hijack Execution Flow: Dynamic Linker Hijacking	2 trials, 7,700 events. Atomics: per-process injection via the `LD_PRELOAD` environment variable (#2), and system-wide injection via `/etc/ld.so.preload` (#1). Only two Linux atomics exist at the pinned ART SHA.
Credential Access		
`T1003.008`	OS Credential Dumping: `/etc/passwd` and `/etc/shadow`	3 trials, 10,999 events. Atomics: read `/etc/shadow` via `cat` as root (#1), read `/etc/passwd` as the unprivileged user (#3), and read both via a non-`cat` binary (#4).
`T1552.001`	Unsecured Credentials: Credentials In Files	3 trials, 97,493 events. Atomics: search the home directory for AWS credential files (#1), recursive `grep` for password-bearing strings across user directories (#3), and search for GitHub credentials in known config paths (#6).
Discovery		
`T1057`	Process Discovery	1 trial, 4,062 events. Atomic: enumerate running processes via `ps` (#1).
`T1082`	System Information Discovery	3 trials, 12,066 events. Atomics: list OS information via `/etc/os-release` and `uname` (#3), discover the hostname via `hostname`/`hostnamectl` (#8), and list loaded kernel modules via `lsmod`/`/proc/modules` (#25).
`T1083`	File and Directory Discovery	3 trials, 154,581 events. Atomics: recursive Unix file-and-directory enumeration via `find`/`ls` (#3 and #4 — two variants of the same atomic family), and SMB/NFS network-share enumeration via `smbclient`/`showmount` (#8). The technique accounts for 21.9% of all attack-event mass.
Command and Control		
`T1105`	Ingress Tool Transfer	3 trials, 11,609 events. Atomics: download-and-execute a remote payload via `curl`/`wget` (#27), exfil-via-download using `whois` as the transport (#14), and `scp` push of a local file to a remote host (#3). Mixed pull and push directions.
Initial Access (paper-defined research scenarios)		
`T1195.002`	Supply Chain Compromise: Compromise Software Supply Chain	*Track A.* 8 trials, 15,703 events. Synthetic npm package whose `postinstall` hook spawns the lineage `npm`→`node`→`sh`→`curl`, reads sandboxed credentials from `~/.fake_secrets/`, base64-encodes them and POSTs them to a loopback listener at `127.0.0.1:8888`, and drops a cron persistence entry that is removed at trial end. Modeled after the Shai-Hulud npm worm [9]; no worm propagation, no real-internet exfil, no real credentials touched.
`CHAIN.CAP01`	Chained post-compromise capstone (paper-defined)	2 trials, 34,130 events. Five Atomic Red Team primitives wrapped in a single labeled interval: T1059.004 atomic 1 (shell spawn) → T1057 atomic 1 (process discovery) → T1003.008 atomic 1 (`/etc/shadow` read) → T1105 atomic 27 (curl-to-loopback ingress) → T1070.004 atomic 1 (`rm` evidence cleanup). Trial 2 (†) is degraded: the T1059.004 step was skipped at script invocation due to a missing trial argument, so the chain ran as a 4-atomic sequence (T1057 → T1003.008 → T1105 → T1070.004).